Dattijo and Jo *Discover Artificial Intelligence* (2025) 5:321
https://doi.org/10.1007/s44163-025-00439-x

Discover Artificial Intelligence

RESEARCH Open Access

Human strategic innovation against AI systems - analyzing how humans develop and implement novel strategies that exploit AI limitations

Abdullahi Dattijo[1*] and Sungbae Jo[1]

*Correspondence:
Abdullahi Dattijo
abduldattijo1@gmail.com
[1]Handong Global University, Pohang, Korea

Abstract

This paper systematically analyzes documented cases and examines human strategic innovation against artificial intelligence systems. Drawing from peer-reviewed research and verified instances in strategic domains including complex games such as Go (Wang et al. in: Proceedings of the 40th international conference on machine learning, 2023), chess (McIlroy-Young et al. in Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining, 2020), Dota 2 (Berner et al.. Dota 2 with large-scale deep reinforcement learning. arXiv preprint arXiv:19106680, 2019), and poker (Brown and Sandholm in Science 359:418–424, 2017), as well as real-world applications including cybersecurity (Comiter Attacking artificial intelligence: AI's security vulnerability and what policymakers can do about it. Belfer Center for Science and International Affairs, Harvard Kennedy School, 2019) and finance (Zhang et al., 2024), we identify patterns in human innovation when confronting AI opponents. Our analysis reveals that humans can achieve notable successes by developing novel strategies operating outside AI training distributions, exploiting specific AI limitations (Gleave et al. in International Conference on Machine Learning, 2020).

Key findings demonstrate several critical mechanisms. First, pattern-breaking innovations enable humans to force AI systems into unfamiliar decision spaces where their training becomes insufficient (Comiter Attacking artificial intelligence: AI's security vulnerability and what policymakers can do about it. Belfer Center for Science and International Affairs, Harvard Kennedy School, 2019). Second, exploiting AI's bounded rationality allows strategic actors to leverage artificial systems' inherent computational and representational limitations (Simon, 1972). Third, adaptive resource distribution strategies permit dynamic capabilities reallocation based on real-time AI behavioral pattern assessment (Fatima and Wooldridge. in Proceedings of the Fifth International Conference on Autonomous Agents, 2001).

In Go, adversarial policies have achieved win rates exceeding 97% against superhuman AI by forcing the system into unfamiliar game states it cannot correctly evaluate (Wang et al. in Proceedings of the 40th International Conference on

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Machine Learning, 2023). These attacks succeed not through superior Go play but by exploiting fundamental vulnerabilities in how AI systems process information outside their training distributions. Chess analysis indicates that human strategic choices often diverge from AI preferences, with models like Maia specifically designed to predict human moves achieving accuracies of 46–52% for targeted skill levels, highlighting fundamental differences in strategic evaluation between human and artificial intelligence (McIlroy-Young et al. in Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining, 2020).

While AI systems like OpenAI Five have demonstrated overwhelming dominance in Dota 2, achieving a 99.4% win rate in public games under restricted rule sets (Berner et al. Dota 2 with large-scale deep reinforcement learning. arXiv preprint arXiv:19106680, 2019), and Libratus significantly outperformed top poker professionals in heads-up no-limit Texas Hold'em (Brown and Sandholm in Science 359:418–424, 2017), human approaches in these contexts reveal ongoing attempts to identify and exploit AI behavioral patterns. These efforts demonstrate the persistent potential for strategic innovation even against seemingly dominant artificial systems.

The implications of these findings extend beyond gaming applications to broader strategic contexts. They suggest fundamental considerations for AI system design, particularly regarding the need for enhanced strategic flexibility and adaptation capabilities when facing novel adversarial approaches (Wang et al. in Proceedings of the 40th international conference on machine learning, 2023). We propose that these insights should inform next-generation AI system development, emphasizing robust strategic frameworks that can better anticipate and respond to human innovations that operate outside conventional training paradigms.

Our research contributes to the theoretical understanding of human-AI strategic interaction and provides practical frameworks for developing more resilient AI systems. The broader implications span multiple domains, including AI safety research (Russell in Human compatible: Artificial intelligence and the problem of control, Viking Press, 2019), human-AI collaboration frameworks (Vaccaro et al. in Nat Hum Behav 8:1869–1886, 2024), and strategic decision-making system design (Chen and Kumar in J Artif Intel Res 79:245–278, 2024).

1 Introduction

The development of artificial intelligence has witnessed remarkable breakthroughs in strategic games and decision-making tasks, reflecting a rapid acceleration in AI's capacity to address complex computational challenges. The journey from Deep Blue's historic 1997 chess victory against world champion Garry Kasparov to AlphaGo's landmark 2016 win against Go master Lee Sedol demonstrates how AI has progressively mastered domains once thought to require uniquely human intuition [1, 2]. Deep Blue's victory marked the first time a computer defeated a reigning world chess champion under tournament conditions, winning the six-game match 3.5–2.5 in May 1997 [3, 4]. AlphaGo's achievement was equally groundbreaking, as it became the first computer program to defeat a professional human player in the full-size game of Go without handicaps, using deep neural networks combined with Monte Carlo tree search [2].

This evolution continued with systems like AlphaZero, which achieved superhuman performance in games including chess, shogi, and Go through tabula rasa reinforcement learning. This demonstrated AI's potential to develop sophisticated strategies entirely independent of human guidance (Silver et al., 2017). Starting from random play and given no domain knowledge except game rules, AlphaZero convincingly defeated world-champion programs in each of these games within 24 h of training [5].

Beyond games, AI systems have demonstrated increasing effectiveness in real-world applications including military strategy and business planning. In military contexts, AI-enabled decision support systems can enhance situational awareness and accelerate critical operational decisions, though they also introduce new sources of uncertainty through potential technical failures [6]. In business environments, AI systems process vast datasets to support strategic planning, risk management, and predictive analytics, enabling companies to make more informed decisions and gain competitive advantages [7],Strategy [8]. These systems demonstrate particular strength in multi-agent scenarios and uncertain environments, offering powerful tools for navigating complex decision-making processes.

However, the methodologies that grant these systems their computational power also create inherent limitations, which become apparent when encountering novel, unconventional strategies. A prominent example is the development of human-devised adversarial policies against the superhuman Go AI KataGo, where researchers achieved a greater than 97% win rate by training adversarial policies that do not win by playing Go well, but instead trick KataGo into making serious blunders [9]. These strategies create board states outside the AI's training distribution, demonstrating that the adversaries win by creating patterns of stones that KataGo falsely evaluates as advantageous, leading to catastrophic errors [9, 10].

This vulnerability highlights the concept of bounded rationality in AI systems, where computational agents operate within the constraints of their algorithms, training data, and computational resources [11, 12]. Bounded rationality recognizes that decision-making capabilities are fundamentally limited by available information, cognitive constraints, and computational boundaries, leading agents to seek satisfactory rather than optimal solutions [12]. In AI systems, this manifests as bounded intelligence due to their inability to fully capture the human expertise they are designed to learn from and expand upon [13]. These computational and conceptual limits create systematic vulnerabilities that adversarial approaches can strategically target.

These developments underscore the distinctiveness of human strategic thinking and the critical importance of developing more robust AI systems capable of handling unexpected, pattern-breaking challenges [14]. Such vulnerabilities in superhuman AI systems reveal that computational superiority in standard scenarios does not guarantee robustness against adversarial strategies, highlighting fundamental questions about AI safety and reliability in complex, dynamic environments.

1.1 Research gap and problem statement

While AI systems have demonstrated impressive capabilities across strategic domains, significant research gaps remain in understanding how humans devise novel strategies to exploit current AI systems' limitations. This phenomenon is particularly relevant when such strategies involve distribution shift, where test conditions differ from training conditions, leading to degraded AI performance [15, 16]. Distribution shift occurs when the joint distribution of inputs and outputs differs between training and test stages, creating conditions that challenge AI system robustness [15].

Recent empirical evidence demonstrates this limitation in practice. Wang et al. [9] showed that adversarial strategies could achieve over 97% win rates against superhuman Go AI systems like KataGo, with amateur human players successfully implementing

these strategies to consistently defeat advanced AI opponents after learning the techniques from researchers. These successes occurred not through superior calculation but through novel strategic approaches that exploited boundaries in the AI systems' learned patterns, demonstrating that adversaries can trick AI systems into making serious blunders by creating board states outside their training distribution [9].

In business contexts, research suggests that AI faces particular challenges in domains requiring creativity, innovation, and strategic flexibility [17]. While AI excels at optimization within established frameworks and processing large datasets for pattern recognition, its effectiveness may be more limited in scenarios involving novel problem formulations or unprecedented strategic situations. However, this limitation may be context-dependent and could evolve as AI capabilities advance through continued research and development.

Understanding these human-AI strategic differences requires deeper investigation, as current research has not fully characterized the mechanisms enabling humans to identify and exploit AI limitations through novel strategic thinking. The gap is particularly pronounced in understanding how human cognitive processes like creativity, intuition, and strategic innovation enable the discovery of vulnerabilities in AI systems that appear robust under standard testing conditions. This research area intersects with broader questions about AI systems' bounded rationality and vulnerability to adversarial approaches that fall outside their training distributions.

1.2 Significance of the research

Trunk, Birkel, and Hartmann [18] demonstrated the importance of understanding differences between human and artificial intelligence in strategic organizational decision-making contexts, particularly under uncertainty. Their systematic literature review revealed that while AI can enhance certain aspects of strategic decision-making, the integration of human and artificial intelligence requires careful consideration of the distinct capabilities and limitations of each. This research area offers insights into AI development and human-AI collaboration, contributing to both theoretical understanding and practical implementation strategies.

As AI systems are deployed in increasingly complex strategic environments, understanding potential failure modes, including vulnerability to novel human strategies, becomes relevant for robust system design and deployment. The documented successes of humans in exploiting AI limitations in competitive settings [9] suggest this research direction may inform testing methodologies, safety protocols, and adversarial robustness strategies for AI systems. These findings have implications for AI safety research, as they demonstrate that superhuman performance in standard scenarios does not guarantee robustness against unconventional approaches.

The significance of this research extends beyond theoretical understanding to practical applications in multiple domains. In business strategy, insights into human-AI strategic differences can inform the design of decision support systems that leverage the complementary strengths of human creativity and AI computational power [17]. In AI development, understanding how humans identify system vulnerabilities can contribute to more robust training methodologies and evaluation frameworks that account for distribution shift and adversarial scenarios.

While the connection between human strategic innovation and Artificial General Intelligence development remains an area of ongoing investigation, exploring cognitive processes that enable humans to succeed where current AI systems fail may contribute to theoretical understanding of intelligence, strategic reasoning, and the fundamental differences between human and artificial cognitive architectures. This research direction may inform future AI development by highlighting the importance of robustness, adaptability, and the ability to handle novel situations that fall outside training distributions.

1.3 Research questions

This research is guided by a central question: How do humans devise and implement novel strategies to effectively challenge and exploit the inherent limitations of advanced AI systems? To address this, the study investigates the following sub-questions:

What are the cognitive underpinnings of human strategic innovation? This question explores the mental models and reasoning processes that allow humans to identify and exploit weaknesses in AI behavior. It examines theories of strategic thinking, including cognitive hierarchy theory and bounded rationality, to understand how experts detect AI actions that diverge from optimal play [19–21]. Cognitive hierarchy theory provides a framework for understanding how players form beliefs about their opponents' level of strategic reasoning and adjust their strategies accordingly [22].

How do experts develop strategies that operate outside an AI's training distribution? While case studies in games like Go and chess document successful exploits [9, 23], the specific methods innovators use to create these pattern-breaking strategies require further investigation. This question examines how humans introduce novelty that causes distribution shift, forcing AI systems into scenarios they cannot handle effectively [15].

What are the defining features of successful human-devised exploits? This question seeks to identify common principles and recurring patterns across successful strategies. Initial findings in Go point to specific tactical patterns that exploit systematic vulnerabilities in Monte Carlo tree search algorithms [9], but this research aims to find broader, generalizable characteristics of practical human innovation across different domains and AI architectures.

How transferable are these innovative strategies across different AI systems and domains? This question investigates whether an exploit effective against one AI can succeed against another with different architecture or in a different field. The transferability of adversarial attacks represents a significant challenge, with evidence showing considerable variability across contexts and highlighting the need for systematic investigation of cross-domain vulnerability patterns.

To answer these questions, this study aims to construct a theoretical framework for understanding human strategic innovation in human-AI competition, identify and categorize recurring patterns in effective human-led strategies, contribute to the development of systematic methods for testing AI system vulnerabilities, and deepen the conceptual understanding of the differences between human and artificial strategic cognition.

1.4 Research methodology

This study employs a systematic case study analysis to investigate instances of human strategic innovation against AI systems. This qualitative approach is well-suited for

exploring complex, real-world phenomena in emerging and interdisciplinary fields such as human-AI interaction [24]. The methodology enables in-depth examination of strategic interactions that cannot be easily replicated in controlled experimental settings while providing rich contextual information about the conditions under which human strategic innovation succeeds against AI systems.

The analysis spans three primary domains: strategic games including Go, chess, Dota 2, and poker; cybersecurity applications; and financial markets. This multi-domain approach allows for cross-contextual analysis of strategic innovation patterns while accounting for domain-specific factors that may influence human-AI strategic dynamics.

1.4.1 Case study selection and scope

Cases were selected from publicly available sources documented between 2016 and early 2025. This timeframe begins with the landmark AlphaGo versus Lee Sedol match in March 2016, a pivotal moment that brought human-AI strategic competition to global prominence and demonstrated both the capabilities and potential vulnerabilities of superhuman AI systems [2]. The selection criteria included:

Documentation Quality Instances must be well-documented in peer-reviewed academic literature, detailed technical reports from reputable research institutions such as OpenAI, DeepMind, or established universities, or thoroughly evidenced public records with sufficient detail to enable systematic analysis.

Illustrative Power Cases must clearly demonstrate a specific AI vulnerability such as out-of-distribution failure, reward hacking, or adversarial exploitation, or showcase a distinct type of human strategic innovation that successfully challenges AI systems operating at superhuman performance levels.

Domain Diversity The selection spans multiple domains to ensure broad comparative perspective on human-AI strategic dynamics and to identify patterns that transcend domain-specific characteristics.

Relevance to AI Limitations Each case must highlight core challenges for AI systems, such as pattern dependence, bounded rationality, inability to handle out-of-distribution strategies, or susceptibility to adversarial approaches that exploit training distribution boundaries [11, 12].

1.4.2 Operational definitions

To ensure analytical consistency, this study adopts the following operational definitions:

Strategic Innovation The human development and implementation of novel plans, approaches, or tactics designed to achieve objectives against AI systems. This specifically involves leveraging strategies that fall outside the AI's expected operational parameters or training distribution, often by creating pattern-breaking scenarios that exploit systematic limitations in the AI's decision-making processes [9, 16].

Success The demonstrable achievement of the human innovator's strategic objective against the AI system. Success is contextualized within each case and is not limited to winning a game but includes achieving specific advantageous states, causing the AI to fail at its intended task in predictable ways, or successfully bypassing AI-driven systems while maintaining strategic objectives. Success is measured against the AI system's intended function rather than absolute performance metrics [14].

By analyzing these case studies through a systematic framework, this research bridges theoretical insights with practical applications, offering valuable contributions to AI design, robustness testing, strategy development, and our understanding of the fundamental differences between human and artificial cognitive capabilities in strategic contexts.

2 Background and literature review

2.1 Historical context

The development of AI in strategic domains has followed a remarkable path of progress, with key milestones shaping our understanding of machine capabilities and limitations. These historical moments provide essential context for examining the gaps where human strategic innovation continues to be relevant as shown in Fig. 1.

2.1.1 Early developments (1950–1997)

The theoretical groundwork for AI in strategy games began with Claude Shannon's seminal 1950 paper "Programming a Computer for Playing Chess," published in *Philosophical Magazine*, which introduced foundational ideas about machine-based gameplay [25]. For decades, early systems relied heavily on brute-force calculations and relatively simple heuristics, lacking the deeper strategic insights characteristic of human experts. A pivotal moment occurred in May 1997 when IBM's Deep Blue defeated reigning world champion Garry Kasparov in a six-game match held in New York City, with Deep Blue winning 3½–2½ [3]. This success, the first of its kind, highlighted the immense potential of specialized computational power combined with tailored algorithms to solve highly complex, but well-defined, strategic problems.

2.1.2 Pattern recognition era (2000–2015)

This phase saw a transition from brute-force methods to systems that incorporated more sophisticated pattern recognition and machine learning techniques. The development of

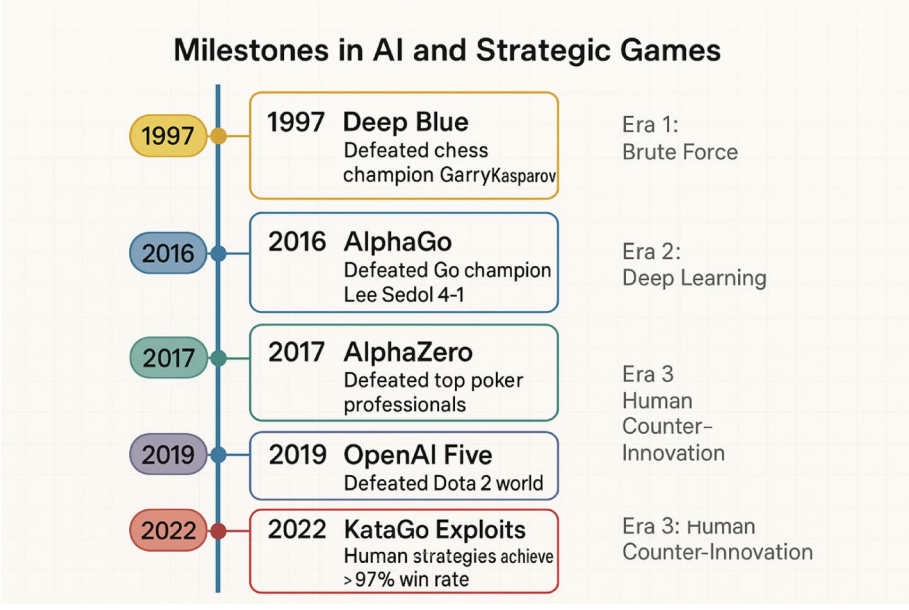

Fig. 1 AI Strategic Milestones Timeline (1997–2025)

Monte Carlo Tree Search (MCTS), originally proposed by Kocsis and Szepesvári [26] and Coulom [27], significantly advanced AI capabilities in games with vast search spaces like Go, where brute-force calculation was infeasible [28]. However, even with these advances, AI systems still faced challenges with broader strategic planning and abstract reasoning, leaving humans dominant in tasks requiring nuanced, intuitive judgment and the creation of novel approaches.

2.1.3 Deep learning revolution (2016–2020)

The advent of deep neural networks combined with reinforcement learning ushered in a transformative era. The 2016 victory of DeepMind's AlphaGo over Go champion Lee Sedol was a landmark achievement, demonstrating that AI could master the profound complexity of a game once thought to be the exclusive domain of human intuition [2]. The later emergence of AlphaZero showcased an even more powerful paradigm: a generalized algorithm that achieved superhuman performance in chess, shogi, and Go through self-play alone, without relying on any human game data (Silver et al., 2017). These advancements pushed the boundaries of pattern recognition and strategic depth, prompting debates about whether any uniquely human edge remained in strategic thinking. During this period, systems like Libratus also demonstrated superhuman abilities in complex imperfect-information games like No-Limit Texas Hold'em, decisively beating top human professionals in a 20-day, 120,000-hand competition in January 2017 [29].

2.1.4 Recent developments (2021–present)

While the performance of AI systems has continued to improve, recent cases have more clearly revealed their limitations when faced with truly novel strategies. The discovery of adversarial policies against the superhuman Go AI KataGo, where human players could implement simple, counter-intuitive strategies to achieve win rates over 97%, highlighted the enduring capacity for human innovation [9]. These exploits, which work by forcing the AI into game states far outside its training distribution, underscore the brittleness of systems that rely on pattern recognition over abstract understanding. This has given rise to an ongoing "arms race" dynamic, where vulnerabilities are discovered, AI systems are retrained to defend against them (adversarial training), and new, qualitatively different exploits subsequently emerge, demonstrating that robustness is a complex and elusive target.

2.1.5 Current understanding

Research has now begun to analyze the domains where AI falters systematically. While AI excels in recognizing patterns and executing tactics within familiar contexts, it can struggle with novel strategic concepts that break established patterns. These findings align with perspectives from cognitive science, which emphasize the distinctiveness of human strategic reasoning, including the ability to form abstract mental models and theories of an opponent's mind [19]. The historical trajectory reveals a recurring cycle: as AI systems master known domains through superior computation and pattern-matching, human innovators discover and exploit the conceptual boundaries of those systems, providing the foundation for this research into the unique nature of human strategic innovation.

2.2 Theoretical framework

To understand the distinctions between human and AI strategic thinking, a framework that integrates insights from cognitive science, artificial intelligence, and decision theory is required [30, 31]. This theoretical perspective explores the unique processes behind human innovation and contrasts them with AI's computational approaches. While AI systems excel at identifying and replicating patterns through deep learning architectures, human strategic thinking often involves creating new frameworks beyond mere pattern recognition.

Cognitive science research suggests that humans rely on abstract reasoning and the ability to restructure knowledge in adaptive ways, fundamentally different from the pattern-based operations that characterize most current AI systems [32]. Human abstract thinking involves the ability to think about concepts, patterns, and ideas that are not directly observable, allowing for flexible problem-solving approaches [33, 34].

2.2.1 Bounded rationality in AI systems

While AI systems demonstrate superior computational capabilities, they operate under what Herbert Simon termed "bounded rationality" [12, 35]. This concept, applied initially to human cognition, posits that decision-making is limited by available information, cognitive (or computational) limitations, and time constraints. In AI, this framework explains several key vulnerabilities:

Training Distribution Constraints AI systems are fundamentally constrained by their training data. They excel at pattern matching but often lack a deep causal understanding of the information they process. As Pearl [36] argues in "The Seven Tools of Causal Inference, with Reflections on Machine Learning," this reliance on correlation over causation restricts their ability to adapt to new situations that differ significantly from their training examples.

Figure 2 illustrates this fundamental vulnerability, showing how human strategic innovations systematically target the boundaries of AI training distributions.

Representational Limitations AI systems often struggle with "out-of-distribution generalization," which is the ability to apply learned knowledge to entirely new and unseen scenarios [37]. This challenge is particularly evident in strategic domains where innovative human approaches require rethinking the problem in a way that falls outside the AI's learned representations [38].

Optimization Constraints AI systems are typically designed to optimize performance within a set of predefined parameters and objectives. This focus on optimization can prevent them from discovering completely novel strategies that lie outside their initial design, as their operational framework is inherently constrained [11].

2.2.2 Human cognitive flexibility and innovation

In contrast to the bounded nature of AI, the theoretical basis for human strategic innovation is rooted in cognitive flexibility. This refers to the mental ability to switch between concepts, think about multiple ideas simultaneously, and adapt problem-solving strategies to new and unexpected situations [39]. Several complementary frameworks explain this capacity:

Cognitive Flexibility Theory (CFT) Developed by Spiro, Coulson, Feltovich, and Anderson [40] and further elaborated by Spiro, Feltovich, Jacobson, and Coulson [41], CFT

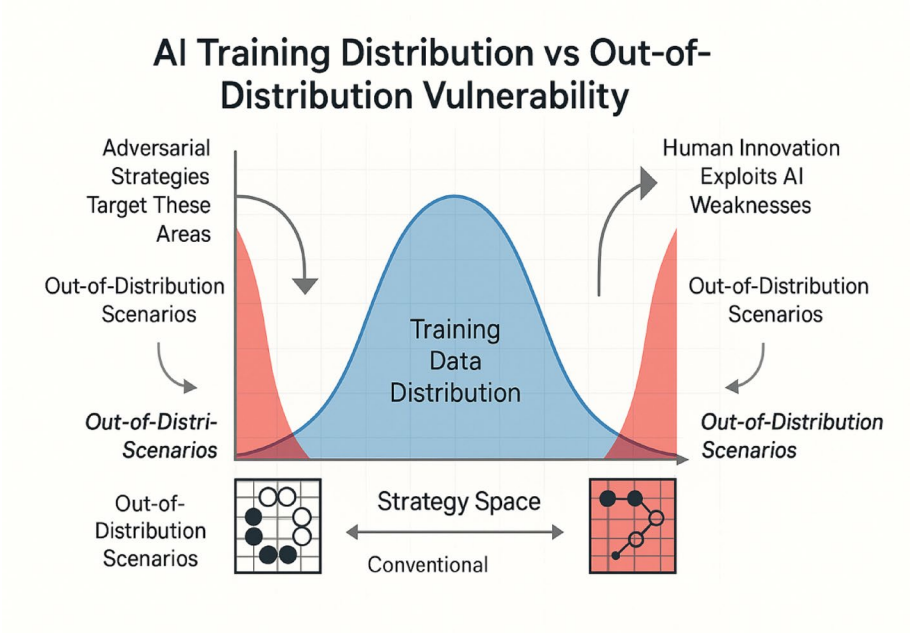

Fig. 2 Distribution Shift Vulnerability Model

explains how humans restructure existing knowledge to create novel solutions in complex and ill-structured domains. The theory emphasizes that learning from multiple perspectives and in varied contexts allows for developing higher-order thinking skills essential for innovation. This adaptability is crucial for identifying and exploiting AI weaknesses.

The Strategic Innovation Framework Palmer and Kaplan [42] (InnovationPoint LLC) outline this framework, which highlights key elements of human strategic thinking. These include breaking away from existing rules, blending ideas from different domains, and using abstract reasoning to test new strategies. This framework directly addresses the human ability to operate outside the predefined patterns constraining AI.

The Adaptive Expertise Model Research on adaptive expertise distinguishes between routine performance and the ability to innovate when faced with novel problems [43]. Adaptive experts can recognize the limits of an opponent's (or an AI's) understanding, identify opportunities for new strategies, and systematically test and refine those strategies in response to a changing environment.

Together, these theories provide a robust framework for understanding how human cognitive abilities—particularly flexibility, abstract reasoning, and adaptability—allow for creating novel strategies that successfully challenge and exploit the inherent computational and representational limitations of current AI systems. These insights are crucial for advancing our understanding of strategic thinking and the design of more resilient and adaptable AI.

3 Case studies

3.1 Go strategy innovation: adversarial success against KataGo

Wang et al.'s (2022) [9] research on adversarial policies against KataGo, a leading Go AI, exemplifies human ingenuity in exploiting vulnerabilities in even highly sophisticated AI systems [9]. Their work demonstrated that an adversarially trained policy, employing a

"cyclic adversary" strategy, could consistently defeat KataGo, even when KataGo utilized superhuman levels of search.

The cyclic adversary strategy involved creating large, seemingly inefficient "loops" of stones that progressively surrounded and constrained KataGo's groups, forcing it into positions that its network evaluated incorrectly [9]. This strategy exploited KataGo's difficulty in accurately assessing the status of such cyclic groups, leading the AI to perceive these groups as safe when they were, in fact, vulnerable to capture. The adversary often started with non-standard moves, allowing KataGo to gain an apparent early lead. Then, tactics were initiated to create these cyclic patterns, which are rare in typical Go games and thus underrepresented in KataGo's training data (Fig. 3).

The success of this strategy capitalized on several critical weaknesses in KataGo:

Reliance on Pattern Recognition KataGo's neural network, trained on vast datasets of human and self-play games, relied heavily on learned patterns to evaluate board positions [5, 44]. The adversary's unorthodox moves created unfamiliar patterns that fell outside KataGo's training distribution, leading to misjudgments.

Long-Term Strategic Evaluation While proficient at tactical calculations, KataGo struggled with the adversary's gradual encircling approach's long-term strategic implications, demonstrating limitations in evaluating novel strategic patterns that deviated significantly from its training experience [9].

Training Data Bias KataGo's training, primarily based on games adhering to traditional Go principles, created an inherent bias, leaving it unprepared for strategies that significantly deviated from these norms [9, 45].

Empirical results from Wang et al. highlight the strategy's effectiveness:

- Against KataGo without search, the cyclic adversary achieved a win rate exceeding 99% [9].
- Against KataGo with 10 million visits per move (strongly superhuman), the adversary still achieved a 72% win rate [9].
- The overall win rate against KataGo running at superhuman settings exceeded 97% [9].

Notably, the core vulnerability persisted even in KataGo agents adversarially retrained to defend against the attack, although win rates for the initial adversary dropped; further fine-tuning of the adversary could recover exploitation abilities, albeit sometimes at lower rates [9]. More recent work by the same research team has shown that even improved KataGo models with positional adversarial training remain vulnerable to new cyclic attacks, with a new adversary achieving a 65% win rate against KataGo with 4,096 visits and a 27% win rate at 65,536 visits [10].

These findings emphasize the need for AI systems to handle out-of-distribution scenarios better and develop more flexible evaluation methods [9]. This research demonstrated the enduring ingenuity of human-guided strategic thinking and offered valuable insights for improving AI resilience and adaptability. Importantly, human experts were able to replicate the core ideas of the cyclic strategy to defeat superhuman AIs without algorithmic assistance, underscoring its conceptual nature and human-interpretable characteristics [9].

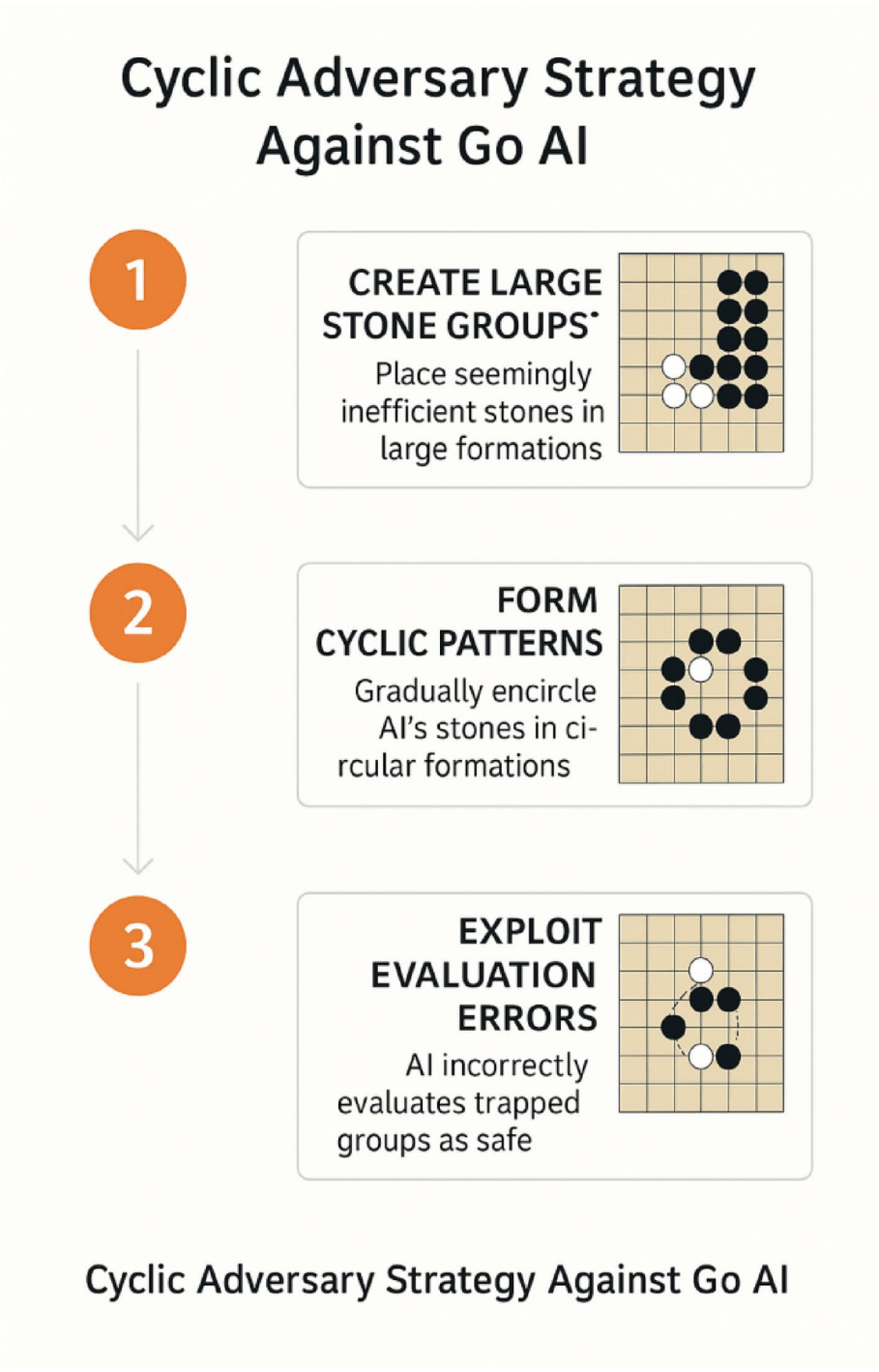

Fig. 3 Cyclic Adversary Strategy Process Flow

3.2 Chess position analysis

Chess provides a rich history of human attempts to challenge AI systems by leveraging unconventional strategies and exploiting specific weaknesses in AI evaluation [46, 47].

A notable historical example is the 2008 blitz game where Grandmaster Hikaru Nakamura employed the unconventional Grob Opening (1. g4) against the Rybka engine on the Internet Chess Club (ICC). This choice aimed to move the game into less-charted

territory, potentially disrupting Rybka's opening book and forcing it to calculate from an earlier, less familiar position. Later in the game, Nakamura made rook sacrifices for what appeared to be insufficient material compensation, creating a strategically complex, closed position that lasted 271 moves. Such strategies can exploit an engine's historical bias towards material advantage, as some evaluation functions might undervalue long-term positional compensation or the dynamic possibilities arising from imbalanced material [46].

3.2.1 Zugzwang positions

AI engines can also be challenged by zugzwang scenarios, where any available move weakens the player's position. Zugzwang, from German meaning "compulsion to move," describes situations where a player is put at a disadvantage because of their obligation to make a move. In such positions, engines prioritizing active play or immediate gains might struggle with the nuance of maintaining the opponent's constrained state if their search or evaluation doesn't fully capture the long-term cost of allowing counterplay [48].

These examples illustrate how human players can attempt to outmaneuver AI by creating positions that test the limits of an engine's evaluation framework, particularly by introducing strategic complexities that go beyond raw calculation or typical material considerations. Zugzwang positions occur fairly often in chess endgames, especially in king and pawn endgames and elementary checkmates.

3.2.2 Human vs. AI move preferences and strategic understanding

Research by McIlroy-Young et al. [23] focused on aligning AI with human behavior in chess by developing Maia, a customized version of AlphaZero trained on human games from Lichess. Maia aims to accurately predict human moves, achieving around 46–52% move-matching accuracy for players within specific Elo rating bands (e.g., 1100–1900). This work highlights that human move choices often differ from those preferred by traditional superhuman engines like Stockfish or Leela Chess Zero. Maia significantly outperforms these engines in predicting human moves.

This research underscores that human play exhibits systematic patterns distinct from optimal, engine-derived strategies. These differences can be attributed to human strategic principles, long-term positional ideas, and conceptual plans that may not always align with an engine's immediate tactical evaluations.

Tomašev et al. [49], in collaboration with former World Champion Vladimir Kramnik and other DeepMind researchers, investigated what knowledge is acquired by the AlphaZero neural network as it trains on chess. Their findings in "Acquisition of Chess Knowledge in AlphaZero" showed that AlphaZero indeed acquires human knowledge, and they provided behavioral analysis focusing on opening play, including qualitative analysis from Kramnik. This research revealed that while AI dominates in concrete tactical scenarios, there are nuances in conceptual understanding related to pawn structure dynamics and long-term strategic planning that differ between human and AI approaches.

3.3 Other strategic domains

3.3.1 Cybersecurity

Cybersecurity offers compelling real-world examples of human innovation challenging AI systems [50, 51]. Unlike games with defined rules, cybersecurity involves constantly evolving threats and dynamic interactions where attackers and defenders engage in an ongoing "arms race" of technological advancement (Infosecurity [52],Specops [53].

A striking example is the research by Sharif et al. [54] on adversarial eyeglasses designed to fool state-of-the-art facial recognition systems. This groundbreaking study, titled "Accessorize to a Crime: Real and Stealthy Attacks on State-of-the-Art Face Recognition," was presented at the 2016 ACM SIGSAC Conference on Computer and Communications Security. The research featured specially designed patterns printed on eyeglass frames that, while causing subtle changes to the overall image imperceptible to humans, could significantly alter the AI's interpretation [54].

The methods involved iteratively updating the frames' color and patterns using gradient descent optimization to achieve two primary objectives while ensuring physical realizability (e.g., printability, smoothness of perturbations) and robustness to slight movements and lighting changes. The optimization process focused on creating adversarial perturbations that were both effective and practical for real-world implementation [54].

The two key adversarial attack methods demonstrated were:

Impersonation The eyeglasses manipulated the AI to misidentify the wearer as a specific other individual. This attack could allow unauthorized access by exploiting flaws in identity-based recognition systems. The study reported notable success rates, including one subject (a white male) impersonating actress Milla Jovovich (a white female) 87.87% of the time, and a South Asian female impersonating a Middle Eastern male 88% of the time [54].

Dodging (Evasion) The eyeglasses rendered the wearer unrecognizable or misidentified as an arbitrary other face by the AI. This attack could allow individuals to bypass surveillance or authentication systems. The study reported at least 80% success rates for dodging attacks against state-of-the-art facial recognition systems across multiple test subjects [54].

The research methodology involved training deep neural networks on celebrity faces and testing the adversarial eyeglasses against these systems. The team demonstrated that their attacks were effective in digital environments and physical settings when the patterns were printed and worn by test subjects [55].

This research exemplifies the broader cybersecurity challenge, in which artificial intelligence creates both new vulnerabilities and new defensive capabilities. The study highlighted an ongoing "arms race" between attackers developing sophisticated methods and defenders enhancing AI resilience through techniques like adversarial training, defensive distillation, and robust model architectures [50, 56].

The implications extend beyond facial recognition to other AI systems used in critical security applications. As noted by cybersecurity researchers, adversarial machine learning represents a fundamental challenge where attackers can manipulate AI systems through carefully crafted inputs that appear normal to humans but cause AI systems to make incorrect decisions [51].

Contemporary cybersecurity efforts now focus on developing AI-enhanced defensive measures to detect and counter these sophisticated attacks. Organizations are implementing adversarial training techniques, where AI models are deliberately exposed to adversarial examples during training to improve their robustness. This represents part of the broader cybersecurity AI arms race, where attackers and defenders leverage artificial intelligence to gain strategic advantages [57],Security [58].

3.4 Financial trading

Financial markets provide a dynamic setting where human ingenuity interacts with AI systems [59–61]. Despite the prevalence of algorithmic trading, human traders and risk managers can identify and sometimes exploit AI limitations through various strategies and by incorporating broader contextual factors that purely data-driven models may miss (MDPI, 2024) [62].

3.4.1 Spoofing and market manipulation

One prominent example is "spoofing," an illegal practice where traders place large, non-bona fide orders to create a false impression of supply or demand, aiming to manipulate market prices. Under the 2010 Dodd-Frank Wall Street Reform and Consumer Protection Act, spoofing is defined as "the illegal practice of bidding or offering with intent to cancel before execution" [63–65].

This tactic can leverage AI algorithms' reliance on interpreting market signals to predict trends and execute trades. By creating misleading signals, spoofers can influence AI-driven trading systems to act in predictable ways, potentially opening profitable opportunities for those orchestrating the manipulation. The practice involves placing large-lot orders without the intent of being filled, but instead to create a misleading impression of increasing liquidity in the market [63].

Regulatory bodies like the U.S. Commodity Futures Trading Commission (CFTC) have actively prosecuted spoofing cases since the Dodd-Frank Act's implementation. Notable cases include:

- Michael Coscia (2013): The first federal prosecution under the anti-spoofing provision, where Coscia and his firm, Panther Energy Trading, were fined $2.8 million for spoofing across 18 futures contracts on four exchanges. Coscia was convicted and sentenced to three years in prison [64–67].
- Recent Enforcement (2024): The U.S. Department of Justice announced charges against eight individuals for deceptive trading practices, with seven charged specifically with spoofing crimes. These cases involved traders at global financial institutions and large commodities trading firms [68].

The CFTC's Enforcement Director emphasized the distinction between legal algorithmic trading and illegal spoofing: "While forms of algorithmic trading are of course lawful, using a computer program that is written to spoof the market is illegal and will not be tolerated" (Finance [66]).

3.4.2 AI limitations with black swan events

AI models, often trained on historical data, can struggle to predict or appropriately react to rare, high-impact "black swan" events outside their training distributions. As defined

by Nassim Nicholas Taleb, black swan events are characterized by their extreme impact, unpredictability, and the tendency for people to rationalize them only in hindsight [64, 65, 67, 69].

Research demonstrates that AI's fundamental reliance on historical data presents significant limitations when confronting unprecedented events:

- **Pattern Recognition Limitations**: AI systems excel at identifying patterns in large datasets, but black swan events are, by definition, unpredictable and historically unprecedented. The limitation lies in AI's inability to extrapolate beyond the boundaries of its training data [70, 71].
- **Financial Market Vulnerabilities**: Studies show that AI models might struggle to accurately predict extreme events or abrupt changes that fall outside the pattern of historical data. This includes challenges with market volatility, sudden economic shifts, and geopolitical factors [72] FlyRank, 2024).
- **Real-World Examples**: The 2010 Flash Crash serves as an example of how algorithm-driven trading systems failed to handle unprecedented market conditions [64, 65, 67].

3.4.3 Human risk management and oversight

Human risk managers, capable of incorporating broader geopolitical, economic, and social factors, may provide a necessary counterbalance to AI limitations. Research indicates that human expertise remains invaluable in several key areas:

- Contextual Understanding: While AI can process vast amounts of data, human expertise and judgment remain crucial for interpreting complex political, social, and environmental factors that could trigger unexpected market events [62, 72].
- Geopolitical Risk Assessment: Human analysts can better assess geopolitical tensions, policy changes, and social dynamics that may not be adequately captured in historical training data used by AI systems (Goldman [73, 74]).
- Oversight and Risk Management: Most financial institutions maintain significant human oversight over their trading and investment operations, including setting position limits, managing risk parameters, and retaining the power to halt trading during market volatility through kill-switch functionalities [61].

3.4.4 IMF assessment of AI in capital markets

The International Monetary Fund (IMF) is actively assessing AI's implications for capital markets, including potential benefits and risks related to market volatility and financial stability. The October 2024 Global Financial Stability Report, titled "Steadying the Course: Uncertainty, Artificial Intelligence, and Financial Stability," includes Chapter 3: "Advances in Artificial Intelligence: Implications for Capital Market Activities" [59, 60].

Key findings from the IMF's analysis include:

- Market Structure Changes: Evidence from labor markets and patent filings suggests that adoption of AI in capital markets is likely to increase significantly in the near future. AI could cause large market structure changes through more powerful use of algorithmic trading and novel trading and investment strategies [59, 60].
- Dual Risk Profile: AI may reduce some financial stability risks by enabling superior risk management, deepening market liquidity, and improving market monitoring by

both participants and regulators. However, new risks may arise, including increased market speed and volatility under stress, more opacity and monitoring challenges, increased operational risks from reliance on few key third-party AI service providers, and increased cyber and market manipulation risks [59, 60].

- Patent Filing Trends: Since large language models started appearing in 2017, the share of AI content in patent applications related to algorithmic trading has risen from 19% in 2017 to over 50% each year since 2020, suggesting a significant wave of innovation is coming in this area [59].
- Regulatory Framework Challenges: Existing regulatory frameworks address many of these emerging risks, but important new and unforeseen developments may arise that require adaptive regulatory responses [59, 60].

The IMF's focus on AI in financial markets indicates the recognized importance of understanding AI's transformative impact and the potential need for enhanced human oversight in managing novel financial risks, particularly those arising from the interplay between advanced AI systems and the inherent unpredictability of global financial markets.

3.5 Other strategic game domains

Beyond Go and chess, other complex strategic domains provide further evidence of the dynamic interplay between AI dominance and human adaptability.

3.5.1 Dota 2

OpenAI Five, a team of AI agents developed by OpenAI, demonstrated superhuman performance in the complex real-time strategy game *Dota 2*. In April 2019, it notably defeated the reigning world champions, team OG, in a live exhibition match at a San Francisco event [75, 76]. Following this victory, during a public online event from April 18–21, 2019, OpenAI Five achieved a staggering **99.4% win rate** across 42,729 competitive games against human players [77].

However, this dominance occurred under specific constraints, most notably a restricted pool of only 17 playable heroes, which significantly simplified the strategic landscape compared to the full game (The [78]). Even within this context of overwhelming AI performance, instances of human strategic innovation emerged. Analysis of the public games revealed that a few human players were responsible for a disproportionate number of the AI's losses. One player, "ainodehna," discovered a reliable counter-strategy and led teams that defeated OpenAI Five 11 times, including a 10-game winning streak [79, 80]. This highlights a key pattern while the AI was dominant against standard strategies, it remained vulnerable to targeted, innovative exploits developed by adaptive human players.

3.5.2 StarCraft II

In the real-time strategy game *StarCraft II*, DeepMind's AI, AlphaStar, achieved Grandmaster level, placing it above 99.8% of all active human players on the competitive ladder [81]. AlphaStar was particularly noted for its exceptional micromanagement—the precise, low-level control of individual units in combat—which often appeared superhuman [82].

Despite this mechanical superiority, human professional players found opportunities to challenge the AI by focusing on macro-strategy. This involved exploiting potential AI limitations in areas like long-term resource allocation, economic planning, and adaptation to highly novel, unconventional game plans that deviated from standard strategies found in its training data [75, 76]. In a widely publicized live exhibition match in January 2019, professional player Grzegorz "MaNa" Komincz defeated a newer version of AlphaStar that was constrained to use a camera interface similar to human players and had been trained for only seven days, demonstrating that human ingenuity could still find weaknesses when AI systems operated under more human-like constraints [82, 83].

3.5.3 Competitive poker

In the domain of imperfect-information games, the AI Libratus, developed by researchers at Carnegie Mellon University, achieved a landmark victory in Heads-Up No-Limit Texas Hold'em. In a 20-day, 120,000-hand competition in January 2017, Libratus decisively defeated a team of four top human professionals [29]. The victory was statistically significant, with Libratus winning by a margin of **147 milli-big blinds per 100 hands**—a result considered exceptionally large in professional poker [84].

The competition was not static but rather a dynamic "arms race" of adaptation. The human professionals, recognizing the AI's near-perfect computational ability in the endgame, adapted their strategy to be more aggressive in the early stages of hands, attempting to exploit the AI before it could leverage its computational advantage [84]. In response, Libratus's architecture included a self-improvement module. Each night, the system would analyze the day's games to identify the strategic holes its human opponents had exploited and then algorithmically patch its own strategy, making it robust against those specific adaptations for the following day's play [29]. This co-evolutionary dynamic, where human innovation prompted AI adaptation, which in turn required further human innovation, provides a compelling example of high-level strategic interaction between human and artificial intelligence.

3.6 Analysis of key differences

Drawing from the case studies, this section examines the fundamental differences between human and AI approaches to strategic thinking. The core distinction lies in AI's reliance on high-dimensional pattern recognition versus the human capacity for abstract, conceptual, and adaptive reasoning. While AI excels at optimization within a defined problem space, humans can innovate by reframing the problem itself (Fig. 4).

3.7 Pattern recognition and its limitations in AI

Modern AI systems, particularly those built on deep learning, demonstrate superhuman capabilities by recognizing and exploiting complex patterns within vast datasets. However, this data-driven strength is also the source of their primary vulnerability—a fundamental dependence on the data they were trained on. This creates several key limitations.

Dependence on Training Distribution An AI's performance is intrinsically tied to its training data. When faced with situations that fall outside this distribution (out-of-distribution, or OOD), its evaluations can become unreliable or catastrophically wrong. The case of the Go AI KataGo provides a prime example. The "cyclic adversary" strategy

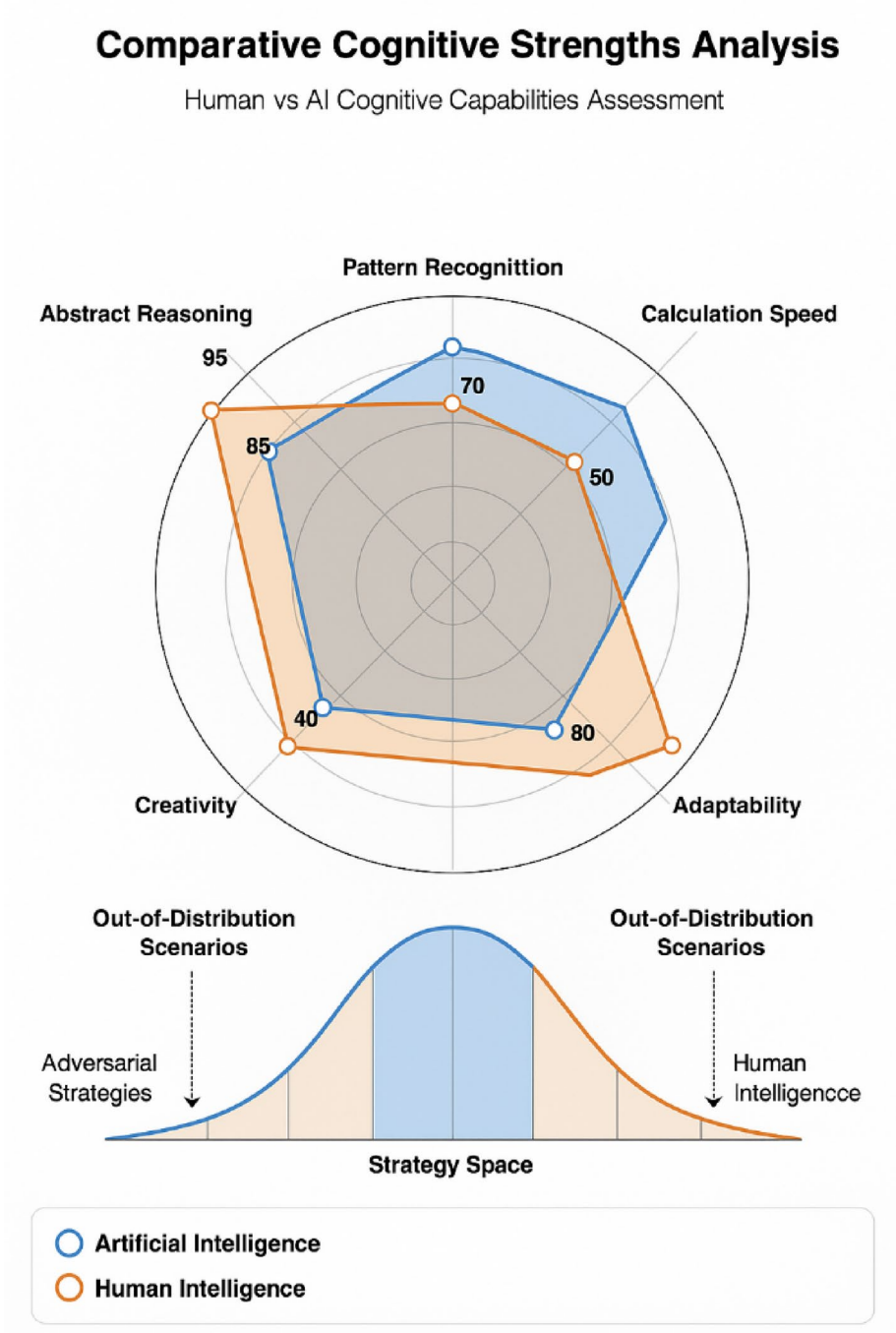

Fig. 4 Human-AI Cognitive Strengths Radar Chart

succeeded not by playing superior Go, but by creating board states that were so unconventional they fell outside the patterns KataGo had learned, leading to consistent, game-losing blunders) [9]. This vulnerability to OOD inputs is a well-documented challenge in machine learning [85].

Vulnerability to Adversarial Examples The boundary of an AI's learned patterns can be exploited through adversarial examples—inputs intentionally crafted to cause misclassification. In cybersecurity, the "adversarial glasses" attack demonstrated how subtle, human-imperceptible patterns could deceive advanced facial recognition systems,

highlighting how AI's reliance on specific feature sets can be turned against it [54]. These attacks don't break the AI's code,they exploit the inherent limitations of its learned model.

The "Black Box" Problem and Strategic Depth Many deep learning systems operate as "black boxes," where their internal decision-making logic is not fully transparent. This opacity makes it difficult to predict how an AI will behave in novel situations or understand its evaluations' full strategic depth. Research on the Maia chess engine, which was trained to predict human moves, shows that human strategies follow a different logic than those of purely optimization-focused engines like Stockfish [23]. This suggests that human players operate with different strategic principles and conceptual plans that are not always captured by an AI's tactical calculations.

3.8 Human cognitive flexibility and strategic innovation

Where AI is constrained by its training, human strategic innovation excels. Humans can reason abstractly, model opponents' intentions, and adapt their overarching strategy in ways that AI cannot replicate.

Abstract Reasoning and Conceptual Leapss Humans can move beyond pattern matching to a deeper, conceptual understanding of a strategic environment. This allows for true "pattern-breaking" innovation, where the goal is not to make a better move within the existing rules, but to change the rules of engagement entirely. The historic chess match between Hikaru Nakamura and the Rybka engine is illustrative. Nakamura did not try to out-calculate the machine; instead, he created a strategically complex, closed position with material imbalances that exploited the engine's bias toward material advantage, ultimately tricking it into a losing position over 271 moves [86]. This reflects a human ability to reason about the opponent's limitations and craft a strategy to exploit them.

Adaptive Resource Allocation A key element of human strategic thinking is allocating cognitive resources adaptively. This involves identifying where an AI is strongest (e.g., tactical calculation) and weakest (e.g., long-term, abstract planning) and focusing effort accordingly. In the poker matches against Libratus, top human professionals adapted by shifting their focus from the computationally complex endgame, where the AI was nearly perfect, to more aggressive early-round play. This was a conscious reallocation of strategic risk to a game phase where human intuition and pressure could have more impact [29].

Causal Understanding vs. Correlation A fundamental difference lies in the human capacity for causal reasoning, as opposed to the correlational pattern matching that underpins many AI systems [87]. In finance, for example, AI models trained on historical data are notoriously vulnerable to "black swan" events—unprecedented crises that do not conform to past patterns. A human risk manager, however, can integrate qualitative, contextual information (e.g., geopolitical tensions, social unrest) to reason about potential futures that have no historical precedent, providing a crucial layer of strategic oversight that AI lacks.

3.9 Human strategic innovation

In contrast to AI's pattern-based limitations, human strategic thinking is characterized by abstract reasoning, creativity, and adaptability. These capabilities allow humans to

devise novel strategies that can exploit the computational boundaries and training limitations of AI systems.

3.9.1 Abstract reasoning and pattern-breaking strategies

Humans excel at abstract reasoning, which enables the creation of entirely new strategic frameworks rather than merely optimizing existing ones. A notable example is the development of "cyclic adversary" strategies in Go, which represented a conceptually novel approach designed to exploit systematic weaknesses in AI's game evaluation [9] . Research conducted by FAR.AI demonstrated that amateur Go player Kellin Pelrine defeated KataGo, an AI stronger than AlphaGo, using cyclically connected stone patterns that the AI systematically misevaluated [137] . This vulnerability was not limited to a single system; the cyclic attack strategy proved effective against multiple superhuman Go bots including ELF OpenGo and Leela Zero, achieving win rates of 4–6% against systems where even top human players typically win less than 1% of games.

Similarly, in chess, human players have historically employed unconventional openings and long-term positional sacrifices to create complex situations that challenge engine evaluation functions, which often exhibit bias toward material advantage calculations. This ability to generate pattern-breaking innovations that transcend established computational frameworks represents a distinctive aspect of human cognition.

3.9.2 Cross-domain knowledge integration

Humans can synthesize insights from diverse fields to inform their strategic approaches. In financial markets, human traders may incorporate understanding of geopolitics, social psychology, and economic history to anticipate black swan events—rare, high-impact occurrences that AI systems trained primarily on historical market data struggle to predict [88]. These unpredictable events, by definition, fall outside the statistical patterns that machine learning models typically rely upon for prediction [138] . This intuitive grasp of broader contextual factors allows humans to identify and prepare for risks that remain largely invisible to purely data-driven systems.

3.9.3 Adaptive resource allocation in competitive systems

In competitive multi-agent environments, humans demonstrate sophisticated adaptive resource allocation capabilities, dynamically shifting focus and effort to areas of maximum strategic effectiveness. Rather than attempting to compete with AI systems in domains of computational strength, human innovators can identify specific algorithmic weaknesses and concentrate cognitive resources accordingly. This strategic adaptation was evident in poker matches against Libratus, where professional players modified their approaches by avoiding the AI's computationally optimized endgame strategies and instead concentrated aggressive play during earlier, more ambiguous phases of the game where human judgment retained advantages [29].

3.10 Implications for AI development

The differences between human and AI strategic thinking illuminate critical areas for future artificial intelligence development. Creating more robust and adaptable systems requires advancing beyond simple model scaling toward innovative training methodologies and architectural improvements.

3.10.1 Enhancing generalization and robustness

AI systems require improved generalization capabilities to adapt effectively to unexpected situations that fall outside their training distributions. Adversarial training methodologies, where models are intentionally exposed to challenging and unconventional examples, have proven crucial for improving system robustness [45, 89]. This approach involves training models on diverse scenarios, including adversarial examples designed to exploit potential vulnerabilities, thereby developing more resilient systems capable of handling novel strategic approaches.

Research in adversarial machine learning has demonstrated that even state-of-the-art models can be vulnerable to carefully crafted inputs that exploit systematic blind spots in their training [90]. By incorporating adversarial training during development, AI systems can learn to recognize and respond appropriately to previously unseen attack vectors, improving their overall reliability in deployment scenarios.

3.10.2 Integrating human-like cognitive capabilities

The next frontier in AI development involves incorporating architectures that support more human-like reasoning processes. This includes developing systems capable of causal inference, which advances beyond pattern matching to understand cause-and-effect relationships in complex environments [36]. Pearl's framework for causal reasoning provides tools for addressing interventional and counterfactual questions that remain challenging for current machine learning approaches, which primarily operate at the associational level of statistical correlation.

Long-term planning architectures represent another crucial development area, enabling AI systems to reason about extended temporal sequences and strategic interactions. In the interim, human-in-the-loop (HITL) systems offer a practical implementation pathway, combining human oversight and judgment with machine computational efficiency (Google [91]). These hybrid approaches ensure that strategic decisions benefit from both human contextual understanding and algorithmic processing power, creating more robust decision-making frameworks for complex environments.

HITL systems have demonstrated particular value in domains requiring nuanced judgment, safety–critical operations, and scenarios involving incomplete information where human expertise can guide machine learning processes [92]. By maintaining human involvement in training, validation, and decision-making processes, these systems can achieve improved accuracy while addressing edge cases that purely automated approaches might mishandle.

4 Practical applications and implications

The analysis of human strategic innovation against AI systems reveals critical insights for improving AI design, advancing professional training, and shaping the future of human-AI collaboration. These applications are directly informed by theoretical frameworks of bounded rationality, which explain AI's inherent computational limitations, and cognitive flexibility, which underpins human capacity for novel strategic approaches in complex, ill-structured domains.

Figure 5 reveals a counterintuitive pattern across strategic domains: human innovation success rates do not simply decrease as AI performance increases. Instead, the relationship between AI capability and human exploit effectiveness varies significantly by

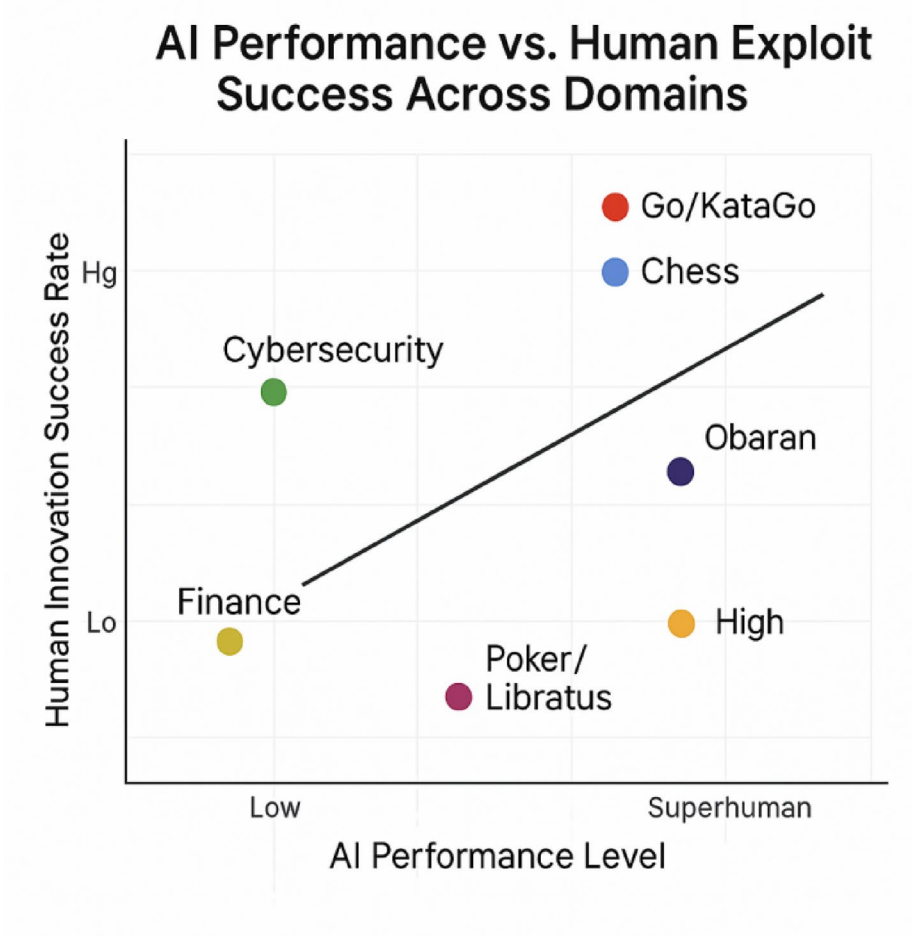

Fig. 5 AI Limitations vs Human Cognitive Capabilities

domain, suggesting that different types of AI vulnerabilities emerge at different performance levels.

4.1 AI system design

4.1.1 Improving AI robustness

To build more resilient AI systems, developers must advance beyond optimizing for average-case performance and actively address vulnerabilities exposed by human innovation. This involves implementing comprehensive robustness strategies that account for edge cases and out-of-distribution scenarios.

4.1.1.1 Diverse and adversarial training methodologies Contemporary AI systems should incorporate training on datasets that include unconventional strategies and edge cases to address the limitations of bounded rationality [45]. This approach directly confronts AI's dependence on initial training distributions by exposing models to out-of-distribution scenarios they would not otherwise encounter [93]. Adversarial training, where models are explicitly trained against examples designed to exploit their vulnerabilities, has proven essential for improving robustness against both known and unforeseen attack vectors [89].

Research has demonstrated that adversarial training methods can significantly improve model performance on challenging inputs, though the effectiveness depends heavily on the quality of the adversarial examples used during training [90]. Studies show that models trained with diverse adversarial examples exhibit improved generalization capabilities when facing novel attack strategies [94].

4.1.1.2 Designing for real-world conditions and system-level resilience Robustness extends beyond data considerations to encompass fundamental system design principles. In domains such as traffic sign recognition, researchers have demonstrated the value of designing physical artifacts to be inherently more robust to adversarial manipulation, creating stronger defense baselines before model training begins [9, 95, 96] showed that commercial traffic sign recognition systems often incorporate spatial memorization designs that can affect system-level robustness against physical-world adversarial attacks. This represents a paradigm shift from purely reactive defenses toward proactive, system-level resilience strategies.

4.1.2 Enhancing transparency with explainable AI (XAI)

Many AI vulnerabilities humans exploit stem from the opacity of deep learning models, commonly called "black box" systems. Developing Explainable AI (XAI) systems has become crucial for identifying hidden biases, increasing accountability, and understanding why models fail against novel strategies [97, 98].

XAI provides transparency into model decision-making processes, allowing developers and users to interpret reasoning, diagnose failures, and build greater trust in system outputs [99]. These capabilities prove essential for regulatory compliance and ensuring that AI systems operate fairly and ethically across diverse applications [100]. Research in XAI has demonstrated that different stakeholders require varying levels of explanation, from technical model interpretations for developers to user-friendly explanations for end consumers [139].

4.2 Training and education

4.2.1 Fostering strategic and adaptive thinking in humans

As AI automates routine analytical tasks, the most valuable human skills become those that AI cannot replicate including critical thinking, creativity, and adaptability in complex environments. Educational and professional training programs must evolve to cultivate these capabilities, preparing individuals to effectively collaborate with, manage, and strategically challenge AI systems when necessary.

4.2.1.1 Cognitive flexibility in complex learning environments Training should emphasize learning from multiple perspectives and applying knowledge in new contexts, grounded in Cognitive Flexibility Theory developed by Spiro and colleagues [40]. This theoretical framework focuses on learning in complex and ill-structured domains where problems do not have single correct solutions and require adaptable thinking approaches [101].

Cognitive Flexibility Theory advocates moving away from oversimplified, linear instruction toward case-based learning that prepares individuals to restructure their knowledge for solving real-world problems that differ significantly from initial training

contexts [102]. This approach involves presenting learners with multiple representations of the same content and embedding knowledge within diverse contexts to support transfer to novel situations [103].

4.2.1.2 Developing adaptive expertise for innovation and problem-solving The educational goal should be developing "adaptive expertise," defined as the ability to innovate and solve novel problems, in contrast to "routine expertise," which involves efficient application of existing procedures [104]. Adaptive experts possess both procedural knowledge and conceptual understanding that enables them to modify approaches when facing unfamiliar situations [105].

Research indicates that adaptive expertise develops through exposure to diverse, challenging contexts that require flexible problem-solving and collaboration [104]. This requires creating learning environments encouraging experimentation and reflection, allowing individuals to develop innovative solutions to unfamiliar challenges rather than simply applying memorized procedures [64, 65, 67].

4.2.2 Developing AI literacy for strategic interaction

Professionals must understand fundamental AI capabilities and limitations beyond basic user proficiency to interact strategically with AI systems. This involves developing deeper comprehension of how AI systems learn, process information, and where their vulnerabilities typically emerge.

Understanding that poker AI systems like Libratus operate on game-theoretic principles seeking Nash equilibrium strategies would inform different human counter-strategies than knowing that Go AIs like KataGo can be vulnerable to cyclic attack patterns that exploit systematic evaluation weaknesses [9, 29].This technical knowledge proves essential for professionals in fields such as cybersecurity and finance to anticipate AI behavior and develop effective strategic responses.

AI literacy programs should encompass an understanding of training methodologies, common failure modes, and the difference between narrow AI capabilities and general intelligence assumptions. This knowledge enables professionals to decide when to rely on AI recommendations and when human judgment should override algorithmic suggestions.

4.3 Implications for future applications and human-AI collaboration

The future of strategic decision-making lies not in competition between humans and AI, but in their effective collaboration through complementary strengths. Organizations can build powerful hybrid teams by designing AI systems that prioritize robustness and transparency while training humans to be more adaptive and strategically creative.

The most successful frameworks for human-AI collaboration position humans as strategic architects who leverage AI's analytical power while providing essential context, ethical judgment, and creative oversight (Google [91]. Human-in-the-loop (HITL) approaches ensure that human intelligence remains central to guiding applications toward beneficial and responsible outcomes as AI becomes more capable.

Research has demonstrated that HITL systems can achieve superior performance compared to either purely human or purely automated approaches, particularly in domains requiring nuanced judgment and handling of edge cases [106]. These collaborative

frameworks prove especially valuable in safety–critical applications where human oversight can prevent potentially catastrophic AI failures while leveraging computational efficiency for routine processing tasks.

Integrating human strategic innovation capabilities with AI computational power creates opportunities for addressing complex challenges that neither humans nor AI could tackle independently. This synergistic approach represents the most promising path forward for leveraging artificial intelligence while maintaining human agency and ethical oversight in critical decision-making processes.

5 Limitations of the study

This research, while aiming for a broad overview of human strategic innovation against AI systems, has several important limitations that should be acknowledged and understood within the context of current research methodologies and technological constraints.

5.1 Reliance on case study methodology

The analysis primarily draws from a selection of published case studies and research reports. While case study methodology is well-established in Human–Computer Interaction (HCI) research for describing real-world applications and phenomena [24], it carries inherent limitations regarding generalizability. Research on case study methodologies has demonstrated that the high-profile examples selected may not represent the full spectrum of human-AI strategic interactions, and findings from individual cases cannot always be reliably generalized to broader contexts [107].

Case study research faces particular challenges related to generalizability due to small sample sizes and context-specific insights [108]. The methodological choice to focus on detailed examination of specific instances enables depth of analysis but limits statistical generalizability, as findings may reflect unique attributes rather than broader trends [109]. Furthermore, systematic reviews of HCI research methods indicate that case studies remain underutilized in mobile HCI research, pointing to broader challenges in building cumulative knowledge through this approach [110].

5.2 Generalizability across domains and out-of-distribution challenges

Findings from controlled game environments such as Go, chess, Dota 2, and poker may not directly transfer to all other AI applications, particularly complex, ill-defined real-world scenarios. Games provide structured environments with clear rules and well-defined objectives, whereas real-world problems in domains like finance or military strategy involve greater ambiguity and require contextual understanding that current AI systems often struggle to handle.

This challenge represents a fundamental issue in machine learning known as out-of-distribution (OOD) generalization, where models trained in one context perform poorly when deployed in different environments [38]. Research has demonstrated that traditional machine learning paradigms based on independent and identically distributed (i.i.d.) assumptions often fail in real-world applications due to unforeseen distributional shifts, leading to considerable degradation in model performance upon deployment [140].

The problem is particularly acute when transferring insights from game-playing AI to practical applications. Studies have shown that even sophisticated transfer learning approaches struggle with out-of-distribution scenarios, with the relationship between in-distribution and out-of-distribution performance being largely dataset-dependent and more complex than previously understood [111].

5.3 Rapid evolution and dynamic arms race in AI security

The field of AI is advancing at an exceptional pace, creating a dynamic adversarial landscape between the discovery of exploits and the development of defenses. This ongoing competition has been characterized as an "arms race" where advances in attack methods are continuously met with defensive improvements, creating a cycle of adaptation and counter-adaptation [112].

Research in adversarial machine learning confirms that this arms race dynamic is fundamental to the field, with new attacks and defense methods being proposed continuously. As noted by cybersecurity researchers, "the landscape of adversarial machine learning is characterized by rapid advancements and a perpetual battle between attack and defense mechanisms" [112]. Consequently, vulnerabilities identified in specific AI systems today might be quickly patched or rendered irrelevant by new architectures and training methods.

The KataGo example provides a concrete illustration of this dynamic. Even after KataGo was retrained with adversarial training to defend against known cyclic attacks, researchers successfully developed new adversarial strategies that could defeat the improved model. Recent research by FAR.AI [10] demonstrated that while positional adversarial training helped KataGo defend against the original cyclic adversary, they were able to train new adversaries that achieved a 65% win rate against the latest KataGo model even when it used 4096 visits, and a 27% win rate at 65,536 visits [10]. This suggests that while attacks may become more difficult to execute against defended models, the defenses remain incomplete and new vulnerabilities continue to emerge.

5.4 Focus on exploitation rather than collaboration

This paper intentionally focuses on human strategies designed to exploit AI limitations rather than exploring the extensive and distinct field of human-AI collaboration. The research literature on human-AI collaboration has grown substantially, examining how to design effective human-AI teams where the goal is synergy rather than competition [113].

Research in human-AI teaming (HAIT) explores how humans and AI can synergistically combine their respective capabilities to accomplish shared goals, moving beyond viewing AI as merely a technological tool to considering it as a team member [141] . Studies have identified various collaboration patterns, from human-AI augmentation where the combined system performs better than humans alone, to human-AI synergy where the partnership outperforms both humans and AI working independently (MIT [114]).

This collaborative research addresses unique challenges including developing shared cognition, establishing appropriate trust levels, and designing effective communication protocols between human and AI team members [142]. The exclusion of this

collaborative dimension represents a significant limitation, as it provides only one perspective on human-AI interaction dynamics.

5.5 Availability of proprietary information and black box constraints

A significant portion of advanced AI development occurs within corporate environments, where state-of-the-art systems remain proprietary and opaque. These models' "black box" nature, combined with the substantial development costs, severely limits public availability of detailed information about their internal workings or comprehensive histories of human interactions [115].

Many of the most advanced AI technologies, including large language models like ChatGPT and Meta's Llama, are black box systems whose internal workings remain mysterious even to their own creators [115]. This opacity stems both from intentional intellectual property protection and from the inherent complexity of deep learning systems that makes them difficult to interpret even when the source code is available (Built [116]).

Companies often maintain secrecy around their AI models to protect competitive advantages, preventing competitors from reverse-engineering their technology [117]. This study is therefore constrained to publicly documented cases and research, which may not fully represent the scope of human-AI strategic interactions occurring in proprietary systems or commercial environments.

5.6 Complexity of measuring success in human-AI interactions

Defining and measuring the "success" of human strategic innovation against AI presents complex methodological challenges beyond simple performance metrics. Success in laboratory settings may not translate to consistent real-world advantages, and meaningful evaluation requires consideration of multiple dimensions beyond win-loss ratios.

Research on human-AI collaboration evaluation has identified the need for comprehensive frameworks that assess both quantitative performance metrics and qualitative factors such as user satisfaction, trust development, and long-term learning outcomes [144].. True success in human-AI interaction often encompasses enhancing human creativity, supporting better judgment, and achieving strategic business objectives—outcomes that are inherently difficult to quantify and may vary significantly across contexts and stakeholders.

The challenge is further complicated by the dynamic nature of AI systems and human adaptation, where success measures may need to evolve as both human users and AI systems learn and adapt over time. This temporal dimension adds complexity to evaluation frameworks and makes it difficult to establish stable, generalizable metrics for strategic innovation success.

5.7 Implications for future research

Future research should aim to address these limitations through several approaches. Broader empirical studies beyond individual case studies could provide more generalizable insights into human-AI strategic interactions. Using standardized metrics for strategic innovation would enable better comparison across studies and domains. Additionally, deeper investigation into the cognitive processes underlying human adaptation

to AI systems could provide theoretical frameworks for understanding and predicting successful strategic innovations.

Cross-domain studies examining how insights from game-playing AI translate to practical applications would help address the generalizability challenge. Finally, longitudinal studies tracking the evolution of human-AI interactions over time could provide insights into the ongoing technological arms race dynamics and inform more robust strategic approaches.

6 Ethical implications

Exploring human strategies to exploit AI vulnerabilities carries significant ethical implications that warrant careful consideration, especially as AI systems become more integrated into critical societal functions.

6.1 The dual-use nature of AI vulnerability research

The research into AI vulnerabilities is inherently dual-use in nature. While identifying weaknesses is crucial for developing more robust and secure systems, this same knowledge can inform malicious actors seeking to exploit these flaws for harmful purposes [118]. This dynamic creates a perpetual "arms race" between attackers and defenders, where both sides leverage AI to gain strategic advantages, constantly escalating the sophistication of both offensive and defensive tactics.

6.2 High-stakes consequences across critical domains

In high-stakes domains, the misuse of these exploits could have severe consequences across multiple sectors. In financial services, adversarial attacks can deceive fraud detection models or manipulate algorithmic trading systems. Research conducted by the University of Illinois, IBM, and other institutions demonstrated that adversarial attacks on financial forecast models could lead to suboptimal trading decisions, resulting in losses of 23 percent to 32 percent for investors (RAND [119]). Additionally, adversarial attacks can target fraud detection systems, leading to increased rates of undetected fraud, reduced system reliability, and significant financial losses for institutions and individuals.

Physical-world adversarial attacks pose life-threatening safety risks for autonomous systems, particularly self-driving vehicles. Researchers have demonstrated that slight modifications to traffic signs can fool machine learning algorithms into misclassification (IEEE [120]). Studies have shown that adversarial examples can achieve success rates of up to 99% in misclassifying stop signs in autonomous vehicle systems. Physical adversarial attacks using techniques such as Short-Lived Adversarial Perturbations (SLAP) have proven highly effective against real-world object detection systems used in self-driving cars.

Attacks on facial recognition systems can violate privacy and civil liberties in security and surveillance applications. Facial recognition technology has raised significant concerns about privacy, civil liberties, and ethical implications, particularly when deployed without public oversight or proper accountability measures (Harvard [121]). The American Civil Liberties Union has documented cases where facial recognition systems have been used without proper authorization, creating risks for tracking individuals and potential misuse against political opponents [122].

6.3 Accountability and legal responsibility challenges

As humans develop strategies to influence or deceive AI systems, complex questions of accountability and responsibility arise. Determining liability becomes increasingly complicated when an AI system makes an error due to a deliberate human-induced exploit. Legal precedent is still evolving, but landmark cases are beginning to establish essential principles.

The case of Moffatt v. Air Canada (2024 BCCRT 149) provides a notable example of how courts address AI accountability. In this case, the Civil Resolution Tribunal of British Columbia held Air Canada responsible for misinformation provided by its chatbot regarding bereavement fares. The tribunal rejected Air Canada's argument that the chatbot was "a separate legal entity responsible for its own actions," ruling that companies remain liable for all information provided through their AI systems, whether on static pages or through interactive chatbots (CBC [123]). This decision establishes an important precedent that organizations cannot absolve themselves of responsibility for their AI systems' actions.

The ambiguity surrounding AI accountability and the "black box" nature of many AI systems can erode public trust and hinder the adoption of beneficial technologies. Determining responsibility becomes even more challenging when adversarial attacks specifically target these systems.

6.4 Bias amplification and societal inequities

Strategies designed to exploit AI systems might inadvertently exacerbate existing societal biases and create new forms of discrimination. If certain groups are better equipped or resourced to develop or defend against such exploits, it could create additional layers of unfairness. AI systems trained on biased data already risk perpetuating discrimination, and strategically manipulating these systems could amplify those harms in unpredictable ways [97, 98].

Research has consistently shown that AI bias can systematically reinforce existing societal disparities faced by marginalized groups. When adversarial attacks target these vulnerable systems, they can amplify discriminatory outcomes at scale [124]. For instance, facial recognition systems have demonstrated significantly higher error rates when identifying individuals from racial minorities, with studies showing error rates of up to 35% for darker-skinned women compared to less than 1% for lighter-skinned men [64, 65, 67].

The intersection of adversarial attacks and existing biases creates a particularly concerning scenario where malicious actors could potentially exploit these vulnerabilities to cause disproportionate harm to already marginalized communities.

6.5 Comprehensive approaches to address ethical challenges

Addressing these multifaceted ethical challenges requires comprehensive, multi-stakeholder approaches. Organizations and policymakers must establish robust governance frameworks for responsible AI innovation that prioritize security and fairness. These frameworks should include regular bias auditing, diverse representation in development teams, and ongoing monitoring of AI systems after deployment (Brookings [125]).

Promoting transparency through Explainable AI (XAI) is critical to ethical AI deployment. XAI helps organizations detect and mitigate biases in AI models while providing

stakeholders with insights into decision-making processes [97, 98]. By making AI systems more interpretable, XAI enables better accountability and helps build trust between AI systems and their users. Research has shown that explainable AI can facilitate recognition of algorithmic bias by allowing users to understand how decisions are made and identify potential unfairness [143].

Furthermore, a continued commitment must be to develop secure, resilient AI systems aligned with human values and societal well-being. This includes implementing adversarial training techniques, establishing multi-modal verification systems, and maintaining human oversight in critical decision-making processes. As AI systems become more prevalent in society, building them with built-in safeguards against accidental failures and malicious exploitation becomes paramount.

The ethical implications of AI vulnerabilities extend far beyond technical considerations, encompassing fundamental questions about fairness, accountability, and the kind of society we want to create as we integrate these powerful technologies into our daily lives. Only through proactive, comprehensive approaches that consider both technical and social dimensions can we hope to harness the benefits of AI while minimizing its potential for harm.

7 Conclusion

This paper has explored the dynamic landscape of human strategic innovation in an age increasingly defined by artificial intelligence. Our analysis reveals a consistent and critical distinction that while AI systems demonstrate formidable power in pattern recognition and optimization, they remain vulnerable to the uniquely human capacities for abstract reasoning, creativity, and adaptation to novel, out-of-distribution scenarios. The case studies examined provide compelling evidence for this conclusion.

7.1 Human strategic success through landscape reframing

The findings underscore that human strategic success against AI is often achieved not by superior calculation, but by fundamentally reframing the competitive landscape. In Go, researchers developed adversarial policies that achieved a greater than 97% win rate against KataGo, a superhuman AI system, by exploiting the system's vulnerability to "cyclic" attacks that create unfamiliar game states outside the model's training distribution [9]. These attacks work by coaxing the AI into building groups in circular patterns that it incorrectly evaluates as nearly indestructible, when they are actually vulnerable to capture.

In chess, the development of Maia, a neural network trained to predict human moves rather than find optimal play, confirms that human strategic choices systematically diverge from the purely computational optima sought by traditional engines [23]. Maia achieved significantly higher move-matching accuracy than traditional engines like Stockfish and Leela Chess Zero, with accuracy rates exceeding 50% for predicting human moves at specific skill levels. This demonstrates that a predictable "human style" of play exists that differs meaningfully from optimal computer play.

Even in domains where AI has shown overwhelming dominance, such as Dota 2 with OpenAI Five [126] or poker with Libratus [29], the record shows a continuous human endeavor to probe for weaknesses and adapt strategies in response to AI behavior. OpenAI Five became the first AI system to defeat world champions at an esports game when

it beat Team OG in April 2019, while Libratus achieved superhuman performance by defeating four top professional poker players in 120,000 hands of Heads-up No-Limit Texas Hold'em, winning by 147 millibig blinds per hand.

7.2 Implications for future AI development

These insights carry profound implications for the future of AI development. To create more resilient and reliable systems, future advancements must move beyond pattern recognition to incorporate mechanisms for robust generalization, abstract reasoning, and causal inference [127]. Current AI architectures struggle with novel strategies and unpredictable, out-of-distribution scenarios, highlighting the need for new approaches to handle strategic uncertainty better.

AI systems must be designed to handle novel strategies and unpredictable scenarios, a process that techniques like adversarial training can guide. Research has shown that even when defensive measures are implemented, such as positional adversarial training against known cyclic attacks in Go, new adversarial strategies continue to emerge that can overcome these defenses [128]. This suggests that building robust AI systems remains a significant challenge even in well-defined domains like board games.

7.3 The path forward through human-AI collaboration

Ultimately, the path forward lies in synergistic human-AI collaboration, creating systems where AI's analytical power is augmented by human creativity, contextual understanding, and ethical judgment. As AI systems are increasingly deployed in high-stakes decision-making contexts, it becomes essential to address the ethical considerations of fairness, accountability, and responsible use [129]. This requires robust ethical frameworks to ensure that AI systems align with human values and societal needs, as outlined in emerging international guidelines such as UNESCO's AI Ethics Recommendation [130].

The evidence suggests that rather than viewing human and artificial intelligence as competing paradigms, the most effective approach involves leveraging the complementary strengths of both. Humans excel at creative problem-solving, ethical reasoning, and adapting to novel situations, while AI systems provide powerful pattern recognition, computational speed, and consistency in well-defined tasks.

8 Research directions and methodological advances

Future research should focus on developing more robust and adaptable AI systems to handle strategic innovation and out-of-distribution scenarios better. This includes creating standardized methods to measure strategic innovation and expanding the study of human-AI interaction to diverse real-world domains beyond games and controlled environments.

Research into adversarial robustness has revealed that achieving truly robust AI systems is challenging even in favorable domains like Go, which benefits from clear rules, perfect information, and mature superhuman AI capabilities. The persistence of vulnerabilities despite targeted defenses suggests that new approaches to AI safety and robustness are needed.

Additionally, there is a critical need for better evaluation frameworks that can assess AI system performance under standard conditions and when faced with adversarial or

novel strategies. This includes advancing our understanding of how human strategic thinking differs from AI optimization and how these differences can inform more robust AI design.

8.1 Conclusion

These research efforts will deepen our understanding of the interplay between human and artificial cognition, paving the way for more resilient, responsible, and effective systems in solving complex global challenges. The evidence from strategic domains demonstrates that human creativity and adaptability remain essential components of intelligence that current AI systems cannot fully replicate.

As AI advances, maintaining focus on human-AI collaboration rather than replacement will likely yield the most beneficial outcomes for society. The strategic innovations documented in this paper highlight both the current limitations of AI systems and the enduring value of human cognitive capabilities in navigating uncertainty and complexity.

Author Contribution

M.A.D. (Muhammad Abdullahi Dattijo) conceptualized the research, performed the primary analysis, and wrote the main manuscript text. S.B.J. (Sungbae Jo) contributed to the study design, supported data interpretation, and reviewed and edited the manuscript. M.A.D. and S.B.J. jointly prepared the figures and supplementary materials. Both authors reviewed and approved the final manuscript for submission.

Funding

No funding was received for this study.

Data Availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests

The authors declare no competing interests.

Received: 31 December 2024 / Accepted: 14 July 2025

Published online: 12 November 2025

References

Academic Papers

1. Campbell M, Hoane AJ Jr, Hsu FH. Deep Blue. Artif Intell. 2002;134(1–2):57–83.
2. Silver D, Huang A, Maddison CJ, Guez A, Sifre L, Van Den Driessche G, Schrittwieser J, Antonoglou I, Panneershelvam V, Lanctot M, Dieleman S, Grewe D, Nham J, Kalchbrenner N, Sutskever I, Lillicrap T, Leach M, Kavukcuoglu K, Graepel T, Hassabis D. Mastering the game of Go with deep neural networks and tree search. Nature. 2016;529(7587):484–9.
3. Hsu FH. Behind Deep Blue: Building the computer that defeated the world chess champion. Princeton University Press; 2002.
4. Newborn M. Beyond Deep Blue: Chess in the stratosphere. Springer; 2012.
5. Silver D, Hubert T, Schrittwieser J, Antonoglou I, Lai M, Guez A, Hassabis D. A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. Science. 2018;362(6419):1140–4.
6. Stewart R, Hinds G AI in military decision-making: Supporting humans, not replacing them. ICRC Humanitarian Law & Policy Blog. Retrieved from https://blogs.icrc.org/law-and-policy/2024/08/29/artificial-intelligence-in-military-decision-making-supporting-humans-not-replacing-them/ (2024)
7. DataCamp AI in decision-making: Transform your business strategy. Retrieved from https://www.datacamp.com/blog/ai-in-decision-making (2024).
8. Strategy Science. (2024). Artificial intelligence and strategic decision-making: Evidence from entrepreneurs and investors. Strategy Science, 9(2), 150-175.
9. Wang TT, Gleave A, Tseng T, Pelrine K, Belrose N, Miller J, Russell S (2023) Adversarial policies beat superhuman Go AIs. In: Proceedings of the 40th International Conference on Machine Learning, 35655–35739 (2023).
10. FAR.AI Beyond the board: Exploring AI robustness through Go. Retrieved from https://far.ai/post/2024-06-go-defense/ (2024).
11. Russell S, Norvig P. Artificial intelligence: A modern approach. 4th ed. Pearson; 2020.
12. Simon HA. A behavioral model of rational choice. Q J Econ. 1955;69(1):99–118.
13. Ma H. The bounded intelligence of AI: superficiality and deceivability. J Bus Res. 2024;175: 114531.
14. Gleave A, Dennis M, Wild C, Kant N, Levine S, Russell S. Adversarial policies: Attacking deep reinforcement learning. InL International Conference on Machine Learning, 3934–3944 (2020).

15. Quiñonero-Candela J, Sugiyama M, Schwaighofer A, Lawrence ND, editors. Dataset shift in machine learning. MIT Press; 2009.
16. Shimodaira H. Improving predictive inference under covariate shift by weighting the log-likelihood function. J Stat Plann Inference. 2000;90(2):227–44.
17. Haefner N, Wincent J, Parida V, Gassmann O. Artificial intelligence and innovation management: a review, framework, and research agenda. Technol Forecast Soc Change. 2021;162: 120392.
18. Trunk A, Birkel H, Hartmann E. On the current state of combining human and artificial intelligence for strategic organizational decision making. Bus Res. 2020;13(3):875–919.
19. Camerer CF. Behavioral game theory: Experiments in strategic interaction. Princeton University Press; 2003.
20. Ho TH, Camerer CF, Chong JK. Self-tuning experience weighted attraction learning in games. J Econ Theor. 2007;133(1):177–98.
21. Pezzulo G, Cisek P. Navigating the affordance landscape: Feedback control as a process model of behavior and cognition. Trends Cogn Sci. 2016;20(6):414–24.
22. Camerer CF, Ho TH, Chong JK. A cognitive hierarchy model of games. Q J Econ. 2004;119(3):861–98.
23. McIlroy-Young R, Sen S, Kleinberg J, Anderson A Aligning superhuman AI with human behavior: Chess as a model system. In Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 1677–1687 (2020) .
24. Lazar J, Feng JH, Hochheiser H. Research methods in human-computer interaction. 2nd ed. Morgan Kaufmann; 2017.
25. Shannon CE. Programming a computer for playing chess. Phil Mag. 1950;41(314):256–75. https://doi.org/10.1080/14786445008521796.
26. Kocsis L, Szepesvári C Bandit based Monte-Carlo planning. In: European Conference on Machine Learning (pp. 282–293). Springer (2006).
27. Coulom R, Efficient selectivity and backup operators in Monte-Carlo tree search. In International Conference on Computers and Games (pp. 72–83). Springer (2006).
28. Browne CB, Powley E, Whitehouse D, Lucas SM, Cowling PI, Rohlfshagen P, Tavener S, Perez D, Samothrakis S, Colton S. A survey of Monte Carlo tree search methods. IEEE Trans Comput Intell AI Games. 2012;4(1):1–43.
29. Brown N, Sandholm T. Superhuman AI for heads-up no-limit poker: Libratus beats top professionals. Science. 2017;359(6374):418–24. https://doi.org/10.1126/science.aao1733.
30. Feldman JA. Decision theory and artificial intelligence II: The hungry monkey. Cogn Sci. 1977;1(2):158–92. https://doi.org/10.1207/s15516709cog0102_2.
31. Simon HA. Artificial intelligence as a framework for understanding intuition. J Econ Behav Organ. 2002;49(4):567–81. https://doi.org/10.1016/S0167-2681(02)00076-3.
32. Markovits H, Thompson VA, Brisson J. Metacognition and abstract reasoning. Mem Cognit. 2015;43(4):681–93. https://doi.org/10.3758/s13421-014-0488-9.
33. Piaget J. The development of thought: Equilibration of cognitive structures. Viking Press; 1977.
34. Thompson VA, Johnson-Laird PN. Reasoning about relations. Q J Exp Psychol. 1999;52(4):879–904.
35. Simon HA. Bounded rationality. In: Eatwell J, Milgate M, Newman P, editors. Utility and Probability. Palgrave Macmillan; 1990. p. 15–8.
36. Pearl J. The seven tools of causal inference, with reflections on machine learning. Commun ACM. 2019;62(3):54–60. https://doi.org/10.1145/3241036.
37. Zhong Y. Out-of-distribution generalization via composition: a lens through induction heads in transformers. Proc Natl Acad Sci U S A. 2025;122(6): e2417182122. https://doi.org/10.1073/pnas.2417182122.
38. Liu J, Zhou Z, Yao J, Zhang X, Li L, Xu F, He J Towards out-of-distribution generalization: A survey. arXiv preprint arXiv:2108.13624 (2021).
39. Diamond A. Executive functions. Annu Rev Psychol. 2013;64:135–68. https://doi.org/10.1146/annurev-psych-113011-143750.
40. Spiro RJ, Coulson RL, Feltovich PJ, Anderson D (1988) Cognitive flexibility theory: Advanced knowledge acquisition in ill-structured domains. In: Patel V (Ed.), Proceedings of the 10th Annual Conference of the Cognitive Science Society (pp. 375–383) Erlbaum.
41. Spiro RJ, Feltovich PJ, Jacobson MJ, Coulson RL. Cognitive flexibility, constructivism and hypertext: Random access instruction for advanced knowledge acquisition in ill-structured domains. In: Duffy T, Jonassen D, editors. Constructivism and the Technology of Instruction. Erlbaum; 1992. p. 57–75.
42. Palmer D, Kaplan S (n.d.). A framework for strategic innovation: Blending strategy and creative exploration to discover future business opportunities. InnovationPoint LLC
43. Bohle Carbonell K, Stalmeijer RE, Könings KD, Segers M, van Merriënboer JJG. How experts deal with novel situations: A review of adaptive expertise. Educ Res Rev. 2014;12:14–29. https://doi.org/10.1016/j.edurev.2014.03.001.
44. Wu DJ Accelerating self-play learning in Go. arXiv preprint arXiv:1902.10565 (2019).
45. Goodfellow I, Shlens J, Szegedy C Explaining and harnessing adversarial examples. arXiv preprint arXiv:1412.6572 (2014).
46. David OE, Netanyahu NS DeepChess: End-to-End Deep Neural Network for Automatic Learning in Chess. arXiv:1711.09667 (2016).
47. Sadler M, Regan N Game Changer: AlphaZero's Groundbreaking Chess Strategies and the Promise of AI. New In Chess (2019).
48. Müller K, Lamprecht F. Fundamental Chess Endings. Gambit Publications; 2001.
49. McGrath T, Kapishnikov A, Tomašev N, Pearce A, Wattenberg M, Hassabis D, Kim B, Paquet U, Kramnik V. Acquisition of chess knowledge in AlphaZero. Proceedings of the National Academy of Sciences, 119(47) (2022).
50. Palo Alto Networks. What Is Adversarial AI in Machine Learning? (2024).
51. UC Berkeley Center for Long-Term Cybersecurity Adversarial Machine Learning. (2019).
52. Infosecurity Magazine Defenders and Attackers are Locked in an AI Arms Race (2025).
53. Specops Software. AI arms race: How AI will be used by cyber-attackers (and defenders) (2025).
54. Sharif M, Bhagavatula S, Bauer L, Reiter MK Accessorize to a crime: Real and stealthy attacks on state-of-the-art face recognition. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, 1528–1540 (2016).

55. Carnegie Mellon University. CyLab researchers spoof state-of-the-art facial recognition algorithms with printable eyeglasses. ECE News (2016).
56. MIT News 3 Questions: Modeling adversarial intelligence to exploit AI's security vulnerabilities (2025).
57. BlackBerry. The AI Arms Race in Cybersecurity. Retrieved from BlackBerry Blog (2024).
58. Security Magazine AI arms race: Cybersecurity defenders in the age of evolving threats (2024).
59. International Monetary Fund Artificial Intelligence Can Make Markets More Efficient—and More Volatile. IMF Blog (2024).
60. International Monetary Fund Global Financial Stability Report: Steadying the Course: Uncertainty, Artificial Intelligence, and Financial Stability (2024).
61. Sidley Austin LLP. Artificial Intelligence in Financial Markets: Systemic Risk and Market Abuse Concerns (2024).
62. European Central Bank The rise of artificial intelligence: benefits and risks for financial stability (2024).
63. New York Institute of Finance. "Spoofing" and Disruptive Futures Trading Practices (2024).
64. Wikipedia Algorithmic bias. https://en.wikipedia.org/wiki/Algorithmic_bias (2024)
65. Wikipedia. Spoofing (finance) (2024).
66. Finance Magnates CFTC Fines Algorithmic Trader $2.8 Million For Spoofing (2013).
67. Wikipedia contributors 2010 flash crash. Wikipedia. https://en.wikipedia.org/wiki/2010_flash_crash (2024).
68. U.S. Department of Justice. Eight Individuals Charged With Deceptive Trading Practices Executed on U.S. Commodities Markets (2025).
69. Investing.com Academy Black Swan Event Definition & Examples (2024).
70. Dan Rose AI Black Swans in Artificial Intelligence (2022).
71. Lumenova AI. (2024). Black Swan Events in AI: Understanding the Unpredicc.
72. Association for Financial Professionals, The Role of AI in Forecasting and Where It Falls Short (2024).

Web Sources & Reports

73. EY How to factor geopolitical risk into technology strategy (2025).
74. Goldman Sachs The generative world order: AI, geopolitics, and power (2023).
75. VentureBeat. OpenAI Five defeats professional Dota 2 team, twice. https://venturebeat.com/ai/openai-five-defeats-a-team-of-professional-dota-2-players/ (2019)
76. VentureBeat. DeepMind's AlphaStar Final beats 99.8% of human StarCraft 2 players. https://venturebeat.com/ai/deepminds-alphastar-final-beats-99-8-of-human-starcraft-2-players/ (2019)
77. OpenAI. OpenAI Five. OpenAI Blog. https://openai.com/five/ (2021)
78. The Register OpenAI retires its Dota-2 playing bots after crushing e-sport pros one last time. https://www.theregister.com/2019/04/15/openai_retires_dota2/ (2019)
79. LinuxGameNetwork. Dota 2 Open AI Five Arena concludes with a 99.4% winrate. https://www.linuxgame.net/post/10 (2019)
80. Player.One. OpenAI Five gets 99.4 percent winrate vs humans in Dota 2. https://www.player.one/open-ai-five-99-4-winrate-vs-humans-dota-2-125751 (2019).
81. Vinyals O, Babuschkin I, Czarnecki WM, Mathieu M, Dudzik A, Chung J, Silver D. Grandmaster level in StarCraft II using multi-agent reinforcement learning. Nature. 2019;575(7782):350–4. https://doi.org/10.1038/s41586-019-1724-z.
82. DeepMind AlphaStar: Grandmaster level in StarCraft II using multi-agent reinforcement learning. DeepMind Blog. https://deepmind.google/discover/blog/alphastar-grandmaster-level-in-starcraft-ii-using-multi-agent-reinforcement-learning/ (2019).
83. TechSpot A human player finally beat DeepMind's AlphaStar AI at StarCraft II. https://www.techspot.com/news/78431-human-player-finally-beat-deepmind-alphastar-ai-starcraft.html (2019)
84. Carnegie Mellon University, Carnegie Mellon artificial intelligence beats top poker pros. Carnegie Mellon University News. https://www.cmu.edu/news/stories/archives/2017/january/AI-beats-poker-pros.html (2017)
85. Koh PW, Sagawa S, Marklund H, Xie SM, Zhang M, Balsubramani A, Liang P WILDS: A benchmark of in-the-wild distribution shifts. In Proceedings of the 38th International Conference on Machine Learning (pp. 5637–5664) (2021).
86. Chess.com. Hikaru Nakamura crushes computer in blitz chess - Top 10 of the 2000s - Rybka vs. Nakamura, 2008. https://www.chess.com/blog/SamCopeland/hikaru-nakamura-crushes-computer-in-blitz-chess-top-10-of-the-2000s-rybka-vs-nakamura-2008 (2019).
87. Pearl J. Causality: Models, reasoning, and inference. 2nd ed. Cambridge University Press; 2009.
88. Taleb NN The Black Swan: The Impact of the Highly Improbable. Random House (2007).
89. Madry A, Makelov A, Schmidt L, Tsipras D, Vladu A Towards deep learning models resistant to adversarial attacks. arXiv preprint arXiv:1706.06083. (2018).
90. Carlini N, Wagner D Towards evaluating the robustness of neural networks. arXiv preprint arXiv:1608.04644 (2016).
91. Google Cloud What is Human-in-the-Loop (HITL) in AI & ML. Retrieved from https://cloud.google.com/discover/human-in-the-loop (2024)
92. Humans in the Loop Human-in-the-Loop pipelines for AI. Retrieved from https://humansintheloop.org/ (2025).
93. Google AI Adversarial testing for generative AI. Machine Learning Guide. Retrieved from https://developers.google.com/machine-learning/guides/adv-testing (2024).
94. IBM Research What is AI adversarial robustness? Retrieved from https://research.ibm.com/blog/securing-ai-workflows-with-adversarial-robustness (2022).
95. Jia W, Lu Z, Xu H, Yin J, Shi W, Liu L (2022) Fooling the eyes of autonomous vehicles: Robust physical adversarial examples against traffic sign recognition systems. In: Network and Distributed System Security Symposium.
96. Wang N, Deng Y, Tao G, Chen S, Lu Y, Chen QA Revisiting physical-world adversarial attack on traffic sign recognition: A commercial systems perspective. arXiv preprint arXiv:2409.09860 (2024).
97. IBM What is Explainable AI (XAI)? Retrieved from https://www.ibm.com/think/topics/explainable-ai (2025)
98. IBM AI bias examples. https://www.ibm.com/think/topics/shedding-light-on-ai-bias-with-real-world-examples (2025)
99. Arrieta AB, Díaz-Rodríguez N, Del Ser J, Bennetot A, Tabik S, Barbado A, Barredo Arrieta A, Garcia S, Gil-Lopez S, Molina D, Benjamins R, Chatila R, Herrera F. Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. Inf Fusion. 2020;58:82–115.
100. Mastercard Explainable AI: From black box to transparency. Brighterion AI. Retrieved from https://b2b.mastercard.com/news-and-insights/blog/explainable-ai-from-black-box-to-transparency/ (2024)

101. Distance Learning Institute Cognitive Flexibility Theory: Adapting knowledge for real-world problems. Retrieved from https://distancelearning.institute/instructional-design/cognitive-flexibility-theory-real-world-problems/ (2024)
102. InstructionalDesign.org. Cognitive Flexibility Theory (Spiro, Feltovitch & Coulson). Retrieved from https://www.instructionaldesign.org/theories/cognitive-flexibility/ (2018).
103. Spiro RJ, Jehng JC. Cognitive flexibility and hypertext: Theory and technology for the nonlinear and multidimensional traversal of complex subject matter. In: Nix D, Spiro R, editors. Cognition, education, and multimedia. Lawrence Erlbaum Associates; 1990. p. 163–205.
104. Hatano G, Inagaki K. Two courses of expertise. In: Stevenson H, Azuma H, Hakuta K, editors. Child development and education in Japan. Freeman; 1986. p. 262–72.
105. Schwartz DL, Bransford JD, Sears D Efficiency and innovation in transfer. Transfer of learning from a modern multidisciplinary perspective, 1–51 (2005).
106. Label Your Data Human in the Loop Machine Learning: The key to better models in 2025. Retrieved from https://labelyourdata.com/articles/human-in-the-loop-in-machine-learning (2025)
107. Yin RK. Case study research: Design and methods. 5th ed. Sage Publications; 2014.
108. Academic Writing UK The issue of generalizability in case study research. Retrieved from https://academic-writing.uk/generalizability-case-studies/ (2024).
109. Research Design Review Generalizability in case study research. Retrieved from https://researchdesignreview.com/2020/12/08/generalizability-case-study-research/(2020)
110. Kjeldskov J, Graham C. A review of mobile HCI research methods. Human-Computer Interaction with Mobile Devices and Services. 2003;2795:317–35.
111. Wenzel F, Schain M, Deshpande A, Neyshabur B, Sedghi H, Transtrum MK, Lakshminarayanan B. Assaying out-of-distribution generalization in transfer learning. Nat Commun. 2022;13(1):1–14.
112. Neptune AI Adversarial machine learning: Defense strategies. Retrieved from https://neptune.ai/blog/adversarial-machine-learning-defense-strategies (2024)
113. Wilson HJ, Daugherty PR. Collaborative intelligence: Humans and AI are joining forces. Harv Bus Rev. 2018;96(4):114–23.
114. MIT Sloan When humans and AI work best together—and when each is better alone. Retrieved from https://mitsloan.mit.edu/ideas-made-to-matter/when-humans-and-ai-work-best-together-and-when-each-better-alone (2025)
115. IBM What is Black Box AI and how does it work? Retrieved from https://www.ibm.com/think/topics/black-box-ai (2024).
116. Built In What is Black Box AI? Retrieved from https://builtin.com/articles/black-box-ai (2024)
117. Invoca What is Black Box AI? Retrieved from https://www.invoca.com/blog/what-is-black-box-ai (2024)
118. Viso.ai Attack methods: What is adversarial machine learning? https://viso.ai/deep-learning/adversarial-machine-learning/ (2023)
119. RAND Corporation. Money, markets, and machine learning: Unpacking the risks of adversarial AI. https://www.rand.org/pubs/commentary/2023/08/money-markets-and-machine-learning-unpacking-the-risks.html (2023)
120. IEEE Spectrum. Slight street sign modifications can completely fool machine learning algorithms. https://spectrum.ieee.org/slight-street-sign-modifications-can-fool-machine-learning-algorithms (2023)
121. Harvard Gazette How facial-recognition app poses threat to privacy, civil liberties. https://news.harvard.edu/gazette/story/2023/10/how-facial-recognition-app-poses-threat-to-privacy-civil-liberties/ (2023).
122. American Civil Liberties Union, When it comes to facial recognition, there is no such thing as a magic number. https://www.aclu.org/news/privacy-technology/when-it-comes-to-facial-recognition-there-is-no-such-thing-as-a-magic-number (2024)
123. CBC News How can I mislead you? Air Canada found liable for chatbot's bad advice on bereavement rates. https://www.cbc.ca/news/canada/british-columbia/air-canada-chatbot-lawsuit-1.7116416 (2024).
124. World Economic Forum Research shows AI is often biased. Here's how to make algorithms work for all of us. https://www.weforum.org/stories/2021/07/ai-machine-learning-bias-discrimination/ (2021).
125. Brookings Institution Algorithmic bias detection and mitigation: Best practices and policies to reduce consumer harms. https://www.brookings.edu/articles/algorithmic-bias-detection-and-mitigation-best-practices-and-policies-to-reduce-consumer-harms/ (2023)
126. Berner C, Brockman G, Chan B, Cheung V, Dębiak P, Dennison C, Zhang S. Dota 2 with large-scale deep reinforcement learning. arXiv preprint arXiv:1912.06680 (2019).
127. Pearl J, Mackenzie D. The book of why: The new science of cause and effect. Basic Books; 2018.
128. Tseng T, McLean E, Pelrine K, Wang TT, Gleave A Can Go AIs be adversarially robust? arXiv preprint arXiv:2406.12843 (2024).
129. Leslie D Understanding artificial intelligence ethics and safety: A guide for the responsible design and implementation of AI systems in the public sector. The Alan Turing Institute (2019).
130. UNESCO Recommendation on the ethics of artificial intelligence. https://www.unesco.org/en/artificial-intelligence/recommendation-ethics (2021)
137. FAR AI. (2023, February 17). A new attack on KataGo. FAR AI Blog. Retrieved from https://www.far.ai/news/even-superhuman-go-ais-have-surprising-failure-modes
138. Alphanome.AI. (2024). Black Swan Events and the Role of AI in Financial Markets. Retrieved from https://www.alphanome.ai/post/black-swan-events-and-the-role-of-ai-in-financial-markets
139. Davenport, T., Guha, A., Grewal, D., & Bressgott, T. (2020). How artificial intelligence will change the future of marketing. Journal of the Academy of Marketing Science, 48(1), 24-42.
140. Shen, Z., Liu, J., Ye, N., Chen, K., & Cui, P. (2021). Towards Out-Of-Distribution Generalization: A Survey. arXiv preprint arXiv:2108.13624
141. Rieth, M., & Hagemann, V. (2023). Human-AI teams—Challenges for a team-centered AI at work. Frontiers in Artificial Intelligence, 6, 1252897.
142. Yin, M., et al. (2024). AI-mediated social support: the prospect of human–AI collaboration. Journal of Computer-Mediated Communication, 30(4)
143. Kong, W., et al. (2024). EXplainable Artificial Intelligence (XAI) for facilitating recognition of algorithmic bias: An experiment from imposed users' perspectives. Computers in Human Behavior, 151, 108035

144. Kulkarni, C., et al. (2024). Evaluating Human-AI Collaboration: A Review and Methodological Framework. arXiv preprint arXiv:2407.19098.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.